KTU 2024 SCHEME · PCECT205 · S2 ECE

Network Theory Playbook

From "what is a node?" to two-port parameters — one friendly step at a time.

INSIDE

M1	Circuit basics, sources, Kirchhoff's laws, mesh & node analysis
M2	Network theorems — superposition to maximum power transfer
M3	Laplace transforms & transient analysis of RL, RC, RLC
M4	Network functions, poles & zeros, two-port parameters
+	Placement & GATE rapid-fire · formula sheet · night-before summary

★ How to use this playbook

Network theory is the grammar of electronics. Every amplifier, filter, power supply and chip interconnect you will meet later is "just" a network — so getting comfortable here pays off for years. The good news: the whole subject runs on a handful of ideas used again and again.

Each topic follows the same rhythm: **plain-words intuition first** (as if you have never seen it before), then the **proper theory** at KTU end-sem depth, then **solved examples** that get gradually harder, and finally an **interview corner** with the questions core-electronics companies and GATE love. Read the intuition even if you "already know" the topic — that is usually where interview answers come from.

Suggested routine: one section a day, solve the examples yourself before reading the solution, and use the end-of-module revision page the night before a test. Every numerical answer in this book has been checked with a quick computer calculation.

The boxes you will see

Key idea — the one thing to remember

Think of it like — an everyday picture

Exam tip — how marks are won

Common mistake — how marks are lost

Interview corner — real placement-style Q&A

Formula — results worth memorising

≡ Contents

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01

Circuit Basics, Mesh & Node Analysis

> *Every circuit question you will ever face — in KTU, GATE or an interview whiteboard — is solved with the tools in this module.*

SYLLABUS CHECKLIST

- ☐ Concept of network and circuit; circuit variables
- ☐ Ideal and practical sources
- ☐ Independent and dependent sources
- ☐ Source transformation
- ☐ Kirchhoff's laws (KCL, KVL)
- ☐ Mesh analysis and node analysis
- ☐ Supermesh and supernode analysis
- ☐ All of the above for DC and AC networks with independent and dependent sources

11 hours · CO1 · Analyse networks using mesh and node methods

1.1 Network, circuit and circuit variables

Imagine you connect a few resistors, a battery and some wires on a breadboard. Any such interconnection of electrical elements is called a **network**. If the interconnection contains at least one **closed path** so that current can actually flow round and come back, we call it a **circuit**. So every circuit is a network, but a network need not be a circuit (an open-ended ladder of resistors with nothing connected is a network, not a circuit).

> **THINK OF IT LIKE**
>
> A road map is a *network* of roads. A *circuit* is a round trip you can actually drive — you leave home and come back. Current, like a car on a round trip, needs a closed loop.

Words you must know

Term	Plain meaning
Node	A point where two or more elements meet. (A long wire with no element on it is still one node.)
Essential node / junction	A node where three or more elements meet — where current actually splits.
Branch	A single element (or a string of series elements) between two nodes.
Loop	Any closed path that does not pass through a node twice.
Mesh	A loop that has no other loop inside it — a "window pane" of a flat (planar) circuit.
Active / passive element	Active elements can deliver net energy (sources). Passive ones (R, L, C) only absorb or store it.
Linear / bilateral	Linear: doubling the cause doubles the effect. Bilateral: behaves the same for current in either direction (R, L, C yes; diode no).
Lumped / distributed	Lumped: each element sits at one place, circuit much smaller than the wavelength. Distributed: e.g. long transmission lines.

Circuit variables

Only a few quantities describe what a circuit is doing:

- **Charge** q (coulomb, C) — the "stuff" that moves.
- **Current** $i = dq/dt$ (ampere, A) — how fast charge moves past a point.
- **Voltage** $v = dw/dq$ (volt, V) — energy given to or taken from each coulomb between two points. Always measured *between* two points.
- **Power** $p = v \cdot i$ (watt, W) and **energy** $w = \int p\, dt$ (joule, J).

KEY IDEA — PASSIVE SIGN CONVENTION

If current **enters the + terminal** of an element, the element **absorbs** power p = vi. If current leaves the + terminal, it **delivers** power. Use this one rule for every element, including sources, and your power balance will always work out: $\Sigma\ p_{absorbed} = 0$ for the whole circuit.

ELEMENT LAWS

$$\text{Resistor: } v = R \cdot i \qquad \text{Inductor: } v = L\,\frac{di}{dt} \qquad \text{Capacitor: } i = C\,\frac{dv}{dt}$$

$$\text{Energy stored: } w_L = \tfrac{1}{2}\,L\,i^2 \qquad w_C = \tfrac{1}{2}\,C\,v^2$$

INTERVIEW CORNER

Q. What is the difference between a node and a junction?
A. Any connection point of two or more elements is a node; a junction (essential node) has three or more elements, so current divides there. Node analysis is usually written at essential nodes.

Q. Why can't the current through an inductor change instantly?
A. $v = L\,di/dt$ — an instant jump in current would need infinite voltage. Equivalently, its stored energy $\tfrac{1}{2}Li^2$ cannot jump. Same logic says capacitor voltage cannot jump.

1.2 Ideal and practical sources

A **source** pushes energy into a circuit. There are two kinds of "promise" a source can make:

- An **ideal voltage source** promises a fixed voltage across its terminals, no matter how much current the load draws.
- An **ideal current source** promises a fixed current through it, no matter what voltage is needed.

Real sources cannot keep these promises forever. A real battery's voltage sags when you draw more current — as if a small resistance R_s were hiding inside, in **series**. A real current source leaks some current internally — as if a resistance R_p sat in **parallel** with it.

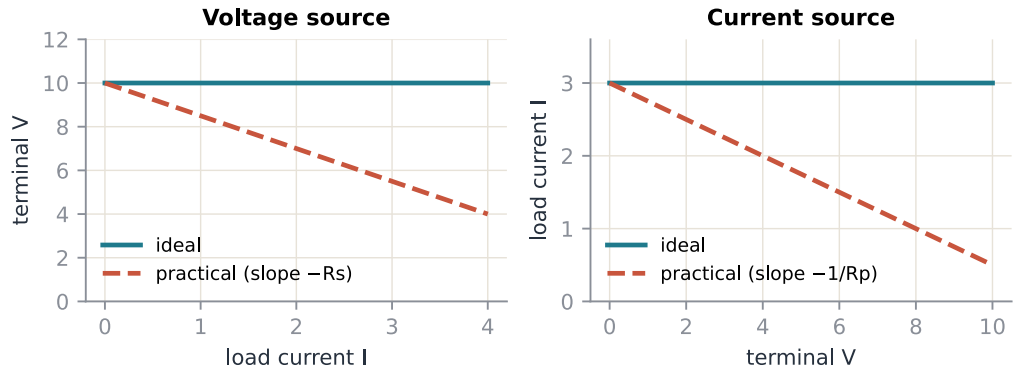


Fig 1.1 Ideal sources give flat lines; practical sources droop because of internal resistance.

PRACTICAL SOURCES

Practical voltage source: $V_{terminal} = V_s - I \cdot R_s$ (ideal when $R_s \to 0$)

Practical current source: $I_{load} = I_s - V/R_p$ (ideal when $R_p \to \infty$)

COMMON MISTAKE

Thinking a current source has zero voltage across it. The voltage across a current source is *whatever the rest of the circuit forces* — you usually have to find it. Likewise the current through a voltage source is unknown until you solve the circuit.

1.3 Independent and dependent sources

An **independent** source has a value set by itself (a 9 V battery, a 2 A source). A **dependent** (controlled) source has a value set by *some other voltage or current in the circuit*. They are drawn as diamonds. Dependent sources are not just exam tricks — they are how we model transistors and op-amps: a BJT's collector current is β times its base current, which is exactly a CCCS.

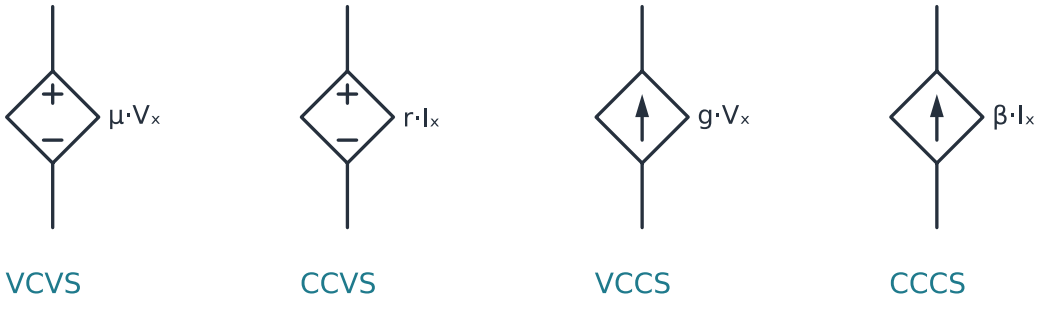


Fig 1.2 The four dependent sources. μ and β are unitless; r is in ohms; g is in siemens.

Type	Output	Controlled by	Constant (unit)	Real-world model
VCVS	voltage μV_x	a voltage	μ (none)	op-amp open-loop gain
CCVS	voltage rI_x	a current	r (Ω)	transresistance amplifier
VCCS	current gV_x	a voltage	g (S)	MOSFET $g_m v_{gs}$
CCCS	current βI_x	a current	β (none)	BJT $\beta \cdot i_b$

KEY IDEA

When solving, treat a dependent source **exactly like an independent one** while writing KCL/KVL, then add one extra equation that expresses its controlling variable (V_x or I_x) in terms of your unknowns. That is the entire trick.

COMMON MISTAKE

"Killing" a dependent source (setting it to zero) like an independent one in superposition or Thevenin resistance. Never do that — a dependent source stays alive as long as its controlling variable can be non-zero.

1.4 Source transformation

A practical voltage source (V_s in series with R_s) and a practical current source (I_s in parallel with R_s) can look **identical from the outside**. Connect any load to terminals a-b of either one and you measure the same voltage and current. So we may swap one for the other whenever it makes a circuit simpler.

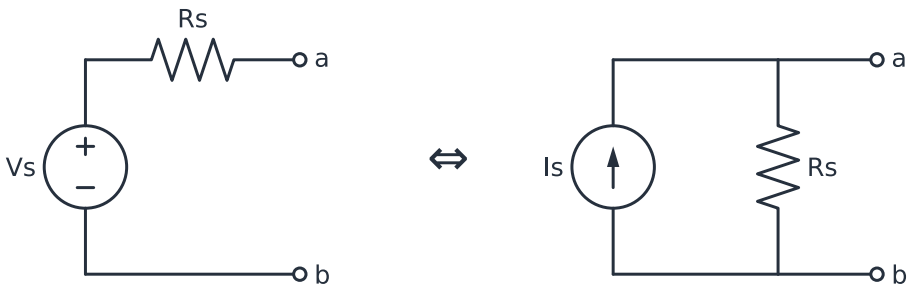


Fig 1.3 Source transformation: the two boxes are equivalent at terminals a-b.

Why it works. Open-circuit the terminals: the left box shows V_s; the right box shows $I_s R_s$. Short them: the left gives V_s/R_s; the right gives I_s. Both match if $I_s = V_s/R_s$. A linear two-terminal box is completely fixed by these two numbers, so the boxes are equivalent.

FORMULA

$$I_s = \frac{V_s}{R_s} \quad \text{(same } R_s\text{; arrow of } I_s \text{ points towards the + terminal of } V_s\text{)}$$

AC: same rule with impedance — $I_s = V_s/Z_s$. Works for dependent sources too, as long as the controlling variable is not destroyed.

COMMON MISTAKE

Trying to transform an *ideal* source. An ideal voltage source has no series R ($R_s = 0$ gives infinite I_s), so it cannot be transformed. Also, the equivalence holds only *outside* the box — the power lost inside R_s is different in the two versions.

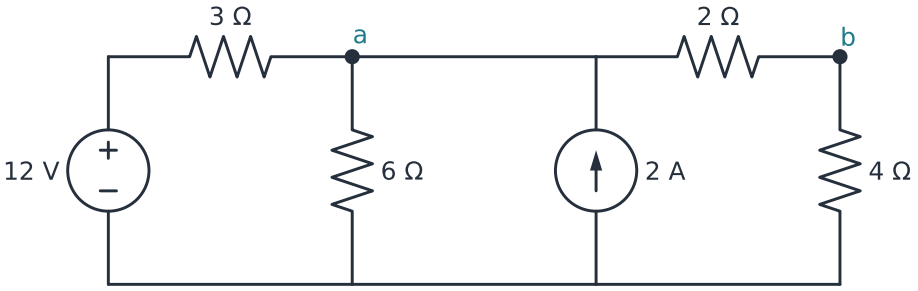


Fig 1.4 Circuit for Example 1.1.

SOLVED EXAMPLE 1.1 · KTU STYLE

Using source transformation, find the current through the 4 Ω resistor in Fig 1.4.

- 12 V in series with 3 Ω → current source 12/3 = 4 A (pointing up into node a) in parallel with 3 Ω.
- Now node a has two current sources (4 A and 2 A, both into a) and two resistors to ground (3 Ω and 6 Ω). Combine: 4 + 2 = 6 A; 3 ∥ 6 = 2 Ω.
- Transform back: 6 A ∥ 2 Ω → 6 × 2 = 12 V in series with 2 Ω.
- The loop is now 12 V, 2 Ω, 2 Ω, 4 Ω in series: I = 12/(2 + 2 + 4) = 1.5 A.

$I_{4\Omega} = 1.5$ A (so $V_b = 6$ V)

1.5 Kirchhoff's laws

Two simple bookkeeping rules let you write equations for any circuit, however tangled.

Kirchhoff's Current Law (KCL)

The algebraic sum of currents at any node is zero: **whatever flows in must flow out**. Charge cannot pile up at a point of wire. KCL is the law of **conservation of charge**.

THINK OF IT LIKE

A water-pipe junction: litres per second coming in equals litres per second going out. A node is just a junction for charge.

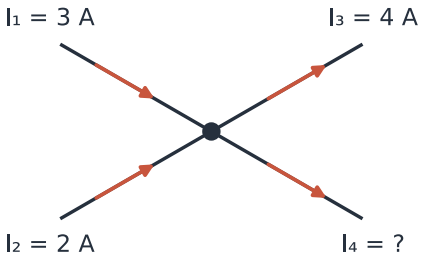


Fig 1.5 KCL at a node.

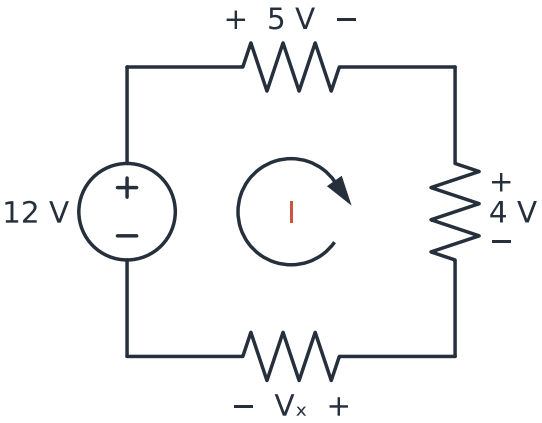


Fig 1.6 KVL around one loop.

Kirchhoff's Voltage Law (KVL)

The algebraic sum of voltages around any closed loop is zero: **total rises = total drops**. If you walk round a loop and return to the start, you are back at the same "height" of potential. KVL is the law of **conservation of energy**.

FORMULA

$$\text{KCL: } \Sigma\, i_{in} = \Sigma\, i_{out} \qquad \text{KVL: } \Sigma\, v \text{ around a closed loop} = 0$$

SOLVED EXAMPLE 1.2 · WARM-UP

(a) In Fig 1.5 find I_4. (b) In Fig 1.6 find V_x.

▸ (a) In: $I_1 + I_2 = 3 + 2 = 5$ A. Out: $I_3 + I_4 = 4 + I_4$. So $I_4 = 1$ A.

▸ (b) Walk clockwise from the source: rise 12 V; drop 5 V across the top; drop 4 V down the right; on the bottom we meet the + end of V_x first, so another drop V_x.

▸ $12 = 5 + 4 + V_x \Rightarrow V_x = 3$ V.

$I_4 = 1$ A, $V_x = 3$ V

EXAM TIP

Before writing a single equation, **mark the polarity** (+/−) of every element and the assumed direction of every current. If a current comes out negative, it simply flows the other way — do not redo the problem.

INTERVIEW CORNER

Q. On what conservation laws are KCL and KVL based?

A. KCL — conservation of charge. KVL — conservation of energy (the electric field in a lumped circuit is conservative, so the work done round a closed path is zero).

Q. When do Kirchhoff's laws fail?

A. When the lumped assumption breaks: at high frequencies where circuit size is comparable to wavelength (distributed circuits, antennas), or when there is a changing magnetic flux linking the loop that is not modelled as an inductor.

1.6 Mesh analysis

Mesh analysis is KVL done smartly. Instead of a current for every branch, we imagine one **circulating current** in each window (mesh) of the circuit. A branch shared by two meshes carries the *difference* of their mesh currents. With fewer unknowns, the algebra shrinks.

THINK OF IT LIKE

Each window of the circuit has its own little conveyor belt running clockwise. Where two belts touch (a shared resistor), they run in opposite directions, so the net flow is the difference of the two belt speeds.

Step-by-step recipe

1. Check the circuit is **planar** (can be drawn with no crossing wires). Count meshes $M = B - N + 1$ (B branches, N nodes).
2. Assign clockwise mesh currents I_1, I_2, … in each window.
3. Write KVL round each mesh: *(own current × total resistance of the mesh) − (neighbour current × shared resistance) = sum of source voltages pushing in the mesh direction.*
4. Add an extra equation for each dependent source's controlling variable.
5. Solve (substitution or Cramer's rule). Branch currents are mesh currents or their differences.

MATRIX FORM BY INSPECTION (INDEPENDENT VOLTAGE SOURCES, ALL CLOCKWISE)

$$[R][I] = [V]$$

R_{kk} = sum of all resistances in mesh k (self-resistance) · R_{kj} = −(resistance shared by meshes k and j) · V_k = net source voltage driving current clockwise in mesh k. [R] is symmetric when there are no dependent sources.

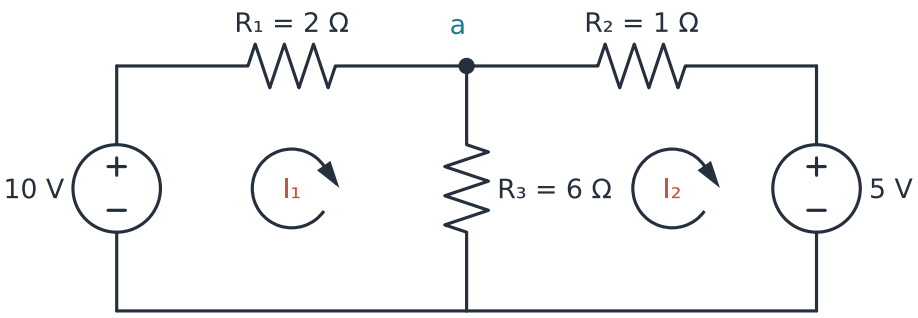


Fig 1.7 Circuit for Example 1.3.

SOLVED EXAMPLE 1.3 · KTU STYLE

Find the mesh currents, the current in R_3 and the voltage at node a in Fig 1.7.

- Mesh 1: $10 = 2I_1 + 6(I_1 - I_2) \Rightarrow 8I_1 - 6I_2 = 10$.
- Mesh 2 (clockwise goes down through the 5 V source from + to −, a drop): $1{\cdot}I_2 + 5 + 6(I_2 - I_1) = 0 \Rightarrow -6I_1 + 7I_2 = -5$.
- $\Delta = 8{\cdot}7 - 36 = 20$. $I_1 = (10{\cdot}7 - (-6)(-5))/20 = 40/20 = 2$ A. $I_2 = (8{\cdot}(-5) - (-6)(10))/20 = 20/20 = 1$ A.
- $I_{R3} = I_1 - I_2 = 1$ A (downward). $V_a = 6 \times 1 = 6$ V. Check: $10 - 2{\times}2 = 6$ V ✓.

$I_1 = 2$ A, $I_2 = 1$ A, $I_{R3} = 1$ A ↓, $V_a = 6$ V

Notice the 5 V source has $I_2 = 1$ A *entering* its + terminal, so it **absorbs** 5 W — it is being charged by the 10 V source.

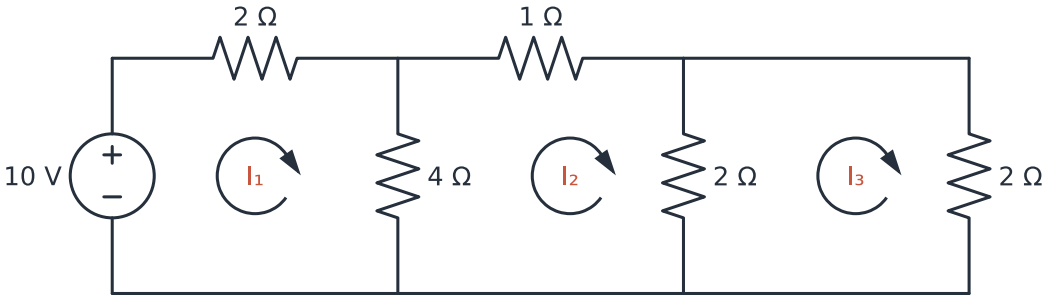


Fig 1.8 Three-mesh ladder for Example 1.4.

SOLVED EXAMPLE 1.4 · MATRIX BY INSPECTION

Write the mesh equations of Fig 1.8 in matrix form by inspection and solve for I_1, I_2, I_3.

- Self-resistances: $R_{11} = 2 + 4 = 6$, $R_{22} = 4 + 1 + 2 = 7$, $R_{33} = 2 + 2 = 4$. Mutual: $R_{12} = -4$, $R_{23} = -2$, $R_{13} = 0$.
- $[6\ {-4}\ 0;\ {-4}\ 7\ {-2};\ 0\ {-2}\ 4]\ [I_1\ I_2\ I_3]^T = [10\ 0\ 0]^T$.
- $\Delta = 6(28 - 4) + 4(-16 - 0) = 144 - 64 = 80$.
- From row 3: $I_2 = 2I_3$. Row 2: $-4I_1 + 14I_3 - 2I_3 = 0 \Rightarrow I_1 = 3I_3$. Row 1: $18I_3 - 8I_3 = 10 \Rightarrow I_3 = 1$ A.

$I_1 = 3$ A, $I_2 = 2$ A, $I_3 = 1$ A

Mesh analysis with a dependent source

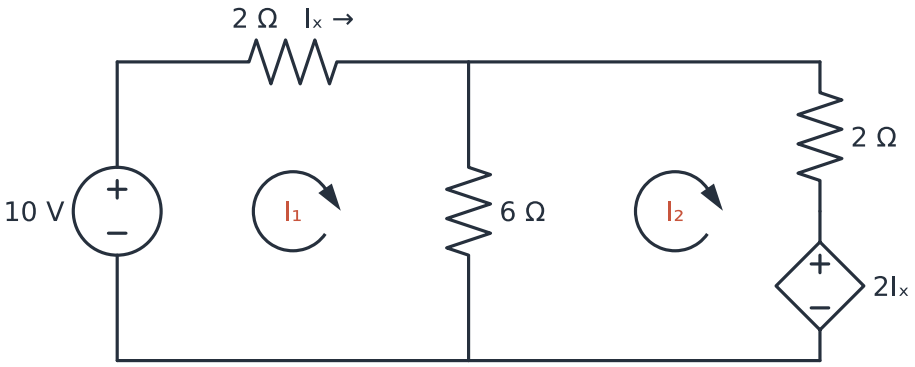


Fig 1.9 Mesh circuit with a CCVS (Example 1.5). I_x is the current in the top-left 2 Ω.

SOLVED EXAMPLE 1.5 · DEPENDENT SOURCE

Find I_1, I_2 and the power absorbed by the dependent source in Fig 1.9.

- Controlling variable: $I_x = I_1$.
- Mesh 1: $10 = 2I_1 + 6(I_1 - I_2) \Rightarrow 8I_1 - 6I_2 = 10$.
- Mesh 2 (clockwise passes down through the CCVS from + to −, a drop of $2I_x$): $2I_2 + 2I_x + 6(I_2 - I_1) = 0 \Rightarrow -4I_1 + 8I_2 = 0 \Rightarrow I_1 = 2I_2$.
- Substitute: $16I_2 - 6I_2 = 10 \Rightarrow I_2 = 1$ A, $I_1 = 2$ A.
- Dependent source: $V = 2I_x = 4$ V; current $I_2 = 1$ A enters its + terminal ⇒ absorbs $4 \times 1 = 4$ W.

$I_1 = 2$ A, $I_2 = 1$ A, $P_{dep} = 4$ W absorbed

EXAM TIP

With a dependent source the [R] matrix is **no longer symmetric** (here −4 and −6 off the diagonal). If a question shows a symmetric matrix for a circuit with a dependent source, someone made a mistake — maybe you!

1.7 Node analysis

Node analysis is KCL done smartly. Pick one node as the **reference** (ground, 0 V). Every other essential node gets an unknown voltage V_1, V_2, … measured from ground. Each branch current is then written as (difference of node voltages)/(resistance). For N nodes you need only N − 1 equations.

THINK OF IT LIKE

Node voltages are heights of water tanks above the floor (ground). Water flows from a higher tank to a lower one, and the flow through a pipe is the height difference divided by the pipe's resistance.

Step-by-step recipe

1. Choose the reference node — usually the one with the most branches (often the bottom wire).
2. Label the other node voltages.
3. At each node write: *sum of currents leaving through every branch* = 0, where a current leaving through R to node j is $(V_k - V_j)/R$, and current sources enter with their own value.
4. Add controlling-variable equations for dependent sources. Solve.

MATRIX FORM BY INSPECTION (INDEPENDENT CURRENT SOURCES)

$$[G][V] = [I]$$

G_{kk} = sum of conductances connected to node k · G_{kj} = −(conductance between nodes k and j) · I_k = net source current *entering* node k.

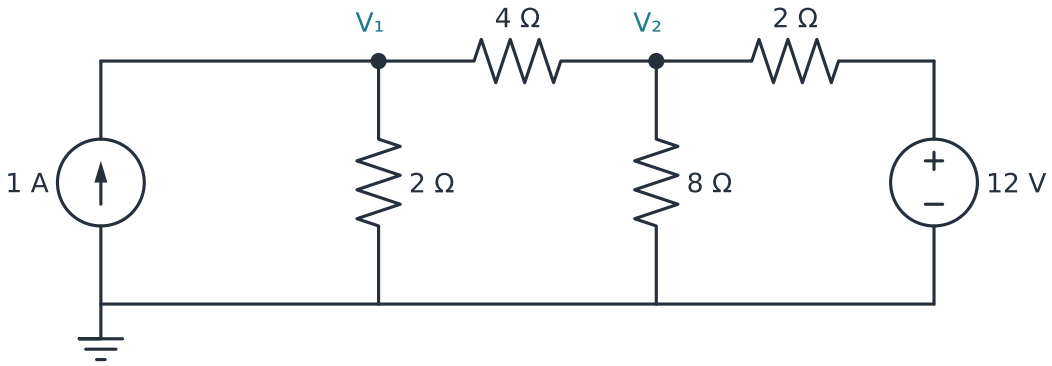


Fig 1.10 Circuit for Example 1.6.

SOLVED EXAMPLE 1.6 · KTU STYLE

Find V_1 and V_2 in Fig 1.10 using node analysis.

▸ Node 1: $\frac{V_1}{2} + \frac{V_1 - V_2}{4} = 1 \Rightarrow 0.75V_1 - 0.25V_2 = 1.$

▸ Node 2: $\frac{V_2 - V_1}{4} + \frac{V_2}{8} + \frac{V_2 - 12}{2} = 0 \Rightarrow -0.25V_1 + 0.875V_2 = 6.$

▸ From node 1: $V_1 = (1 + 0.25V_2)/0.75$. Substituting: $0.875V_2 - (1 + 0.25V_2)/3 = 6 \Rightarrow 2.625V_2 - 1 - 0.25V_2 = 18 \Rightarrow V_2 = 8$ V.

▸ $V_1 = (1 + 2)/0.75 = 4$ V. Check node 1: $4/2 + (4 - 8)/4 = 2 - 1 = 1$ A ✓.

$V_1 = 4$ V, $V_2 = 8$ V

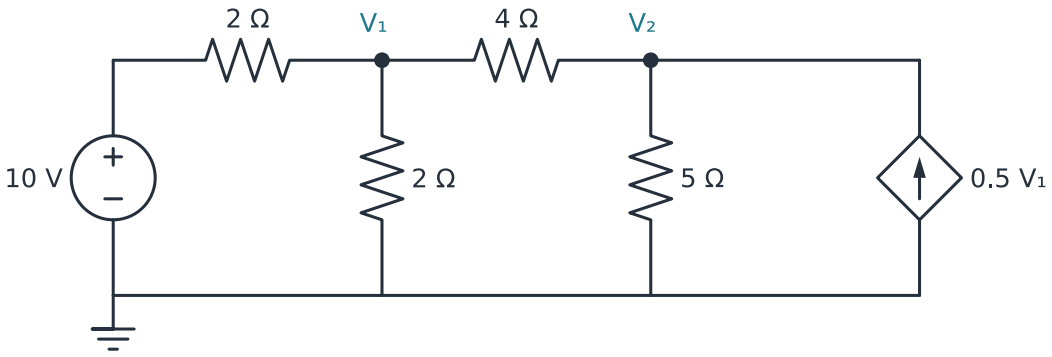


Fig 1.11 Node circuit with a VCCS (Example 1.7).

SOLVED EXAMPLE 1.7 · DEPENDENT SOURCE

In Fig 1.11 the VCCS delivers $0.5V_1$ amperes into node 2. Find V_1, V_2 and the power supplied by the dependent source.

- Node 1: $(V_1 - 10)/2 + V_1/2 + (V_1 - V_2)/4 = 0 \Rightarrow 1.25V_1 - 0.25V_2 = 5$.
- Node 2: $(V_2 - V_1)/4 + V_2/5 = 0.5V_1 \Rightarrow -0.75V_1 + 0.45V_2 = 0 \Rightarrow V_2 = (5/3)V_1$.
- Substitute: $1.25V_1 - (5/12)V_1 = 5 \Rightarrow (10/12)V_1 = 5 \Rightarrow V_1 = 6$ V, $V_2 = 10$ V.
- Source current $0.5 \times 6 = 3$ A, pushed up into node 2 at 10 V ⇒ supplies $3 \times 10 = 30$ W.

$V_1 = 6$ V, $V_2 = 10$ V, $P_{dep} = 30$ W supplied

MESH OR NODE — WHICH ONE?

Count the equations. Mesh needs B − N + 1 equations; node needs N − 1. Choose the smaller. Lots of current sources → node is easier; lots of voltage sources → mesh is easier. A non-planar circuit (wires must cross) can only be done by node analysis.

1.8 Supermesh and supernode

Supermesh — a current source shared by two meshes

KVL needs the voltage across every element, but the voltage across a current source is unknown. So when a current source sits on the boundary between two meshes, we **avoid it**: draw a loop around both meshes, skipping the branch with the current source. That combined loop is a **supermesh**. We then write:

- **one KVL** around the supermesh, and
- **one constraint**: the current source equals the difference of the two mesh currents.

(A current source on the *outer* edge of a single mesh is even easier: that mesh current simply equals the source value.)

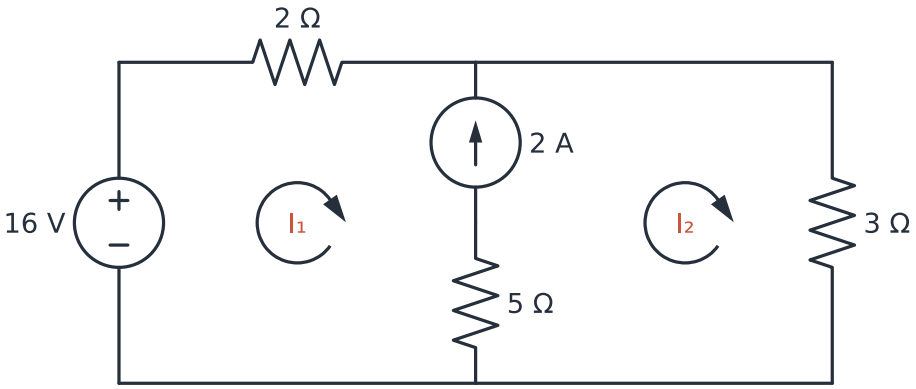


Fig 1.12 Supermesh circuit for Example 1.8.

SOLVED EXAMPLE 1.8 · SUPERMESH

Find the mesh currents in Fig 1.12. Does the 5 Ω resistor affect them?

▸ The 2 A source (pointing up) is in the middle branch. In that branch I_2 flows up and I_1 flows down, so the constraint is $I_2 - I_1 = 2$.

▸ Supermesh KVL round the outside (16 V, 2 Ω, 3 Ω), skipping the middle branch: $16 = 2I_1 + 3I_2$.

▸ Substitute $I_2 = I_1 + 2$: $16 = 5I_1 + 6 \Rightarrow I_1 = 2$ A, $I_2 = 4$ A.

▸ The 5 Ω is in series with an ideal current source, so it never enters the equations — it changes only the voltage across the current source (here 22 V), not any current.

$I_1 = 2$ A, $I_2 = 4$ A; 5 Ω has no effect on the currents

Supernode — a voltage source between two non-reference nodes

KCL needs the current in every branch, but the current through a voltage source is unknown. If a voltage source connects two non-reference nodes, we draw a bubble around both nodes and the source — a **supernode** — and write:

- **one KCL** for the whole bubble (currents leaving the bubble sum to zero), and
- **one constraint**: the difference of the two node voltages equals the source voltage.

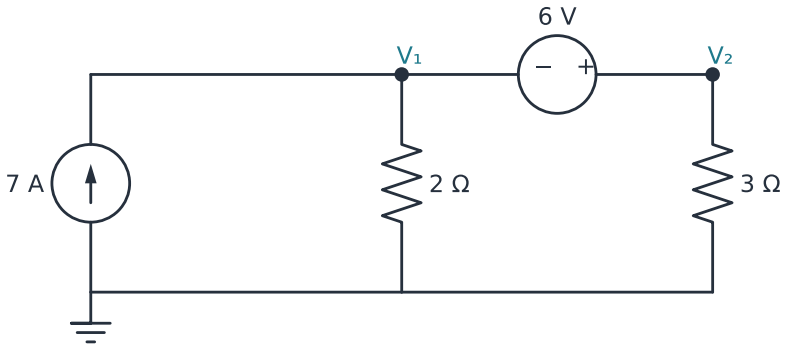


Fig 1.13 Supernode circuit for Example 1.9.

SOLVED EXAMPLE 1.9 · SUPERNODE

Find V_1, V_2 and the current in each resistor of Fig 1.13.

- Constraint from the 6 V source (+ on the V_2 side): $V_2 - V_1 = 6$.
- KCL for the supernode $\{V_1, V_2\}$: 7 A enters; $V_1/2$ and $V_2/3$ leave $\Rightarrow V_1/2 + V_2/3 = 7$.
- Multiply by 6: $3V_1 + 2(V_1 + 6) = 42 \Rightarrow 5V_1 = 30 \Rightarrow V_1 = 6$ V, $V_2 = 12$ V.
- $I_{2\Omega} = 3$ A, $I_{3\Omega} = 4$ A, total 7 A ✓.

$V_1 = 6$ V, $V_2 = 12$ V, $I_{2\Omega} = 3$ A, $I_{3\Omega} = 4$ A

COMMON MISTAKE

Writing a KCL at each of the two nodes inside a supernode separately (that introduces the unknown source current) *and* also the supernode KCL — this double-counts. Use either: supernode KCL + constraint, nothing else for those two nodes.

INTERVIEW CORNER

Q. What happens to the supernode idea if the voltage source is connected to the reference node?

A. No supernode is needed — that node's voltage is simply known (equal to the source voltage), so you have one fewer unknown.

Q. Can you use mesh analysis on a non-planar circuit?

A. No — meshes are defined only for planar circuits. Use node analysis (or general loop analysis with a tree/co-tree).

1.9 Mesh and node analysis of AC networks

For sinusoidal steady state nothing new is needed: replace every element by its **impedance** and every source by its **phasor**, then do exactly the same mesh/node steps using complex numbers.

PHASOR IMPEDANCES (Ω = 2ΠF)

$$Z_R = R \qquad Z_L = j\omega L \qquad Z_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C}$$

Admittance $Y = 1/Z$ (siemens). Average power $P = \mathrm{Re}(V \cdot I^*) = |V||I| \cos\theta$ (with peak phasors, multiply by ½).

KEY IDEA

j means "rotate by 90°". An inductor's voltage leads its current by 90° (+j); a capacitor's voltage lags by 90° (−j). Everything else — KVL, KCL, mesh, node, supermesh, dependent sources — carries over unchanged.

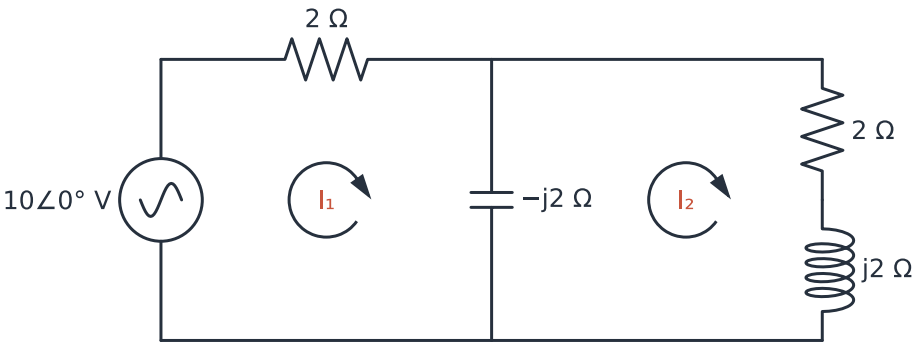


Fig 1.14 AC mesh circuit for Example 1.10 (impedances in ohms).

SOLVED EXAMPLE 1.10 · AC MESH

Find I_1, I_2, the capacitor current and the average power delivered by the source in Fig 1.14 (rms phasors).

- Mesh 1: $(2 - j2)I_1 - (-j2)I_2 = 10 \Rightarrow (2 - j2)I_1 + j2\, I_2 = 10$.
- Mesh 2: $-(-j2)I_1 + (-j2 + 2 + j2)I_2 = 0 \Rightarrow j2\, I_1 + 2I_2 = 0 \Rightarrow I_2 = -jI_1$.
- Substitute: $(2 - j2)I_1 + j2(-jI_1) = (2 - j2 + 2)I_1 = (4 - j2)I_1 = 10 \Rightarrow I_1 = 10/(4 - j2) = 2 + j1$ A.
- $I_1 = 2.236\angle 26.57°$ A; $I_2 = -j(2 + j1) = 1 - j2 = 2.236\angle -63.43°$ A.
- $I_C = I_1 - I_2 = 1 + j3 = 3.162\angle 71.57°$ A (downward).
- $P = \mathrm{Re}(10 \times I_1^*) = \mathrm{Re}(10(2 - j1)) = 20$ W. Check: $2|I_1|^2 + 2|I_2|^2 = 10 + 10 = 20$ W ✓.

$I_1 = 2.236\angle 26.57°$ A, $I_2 = 2.236\angle -63.43°$ A, $I_C = 3.162\angle 71.57°$ A, $P = 20$ W

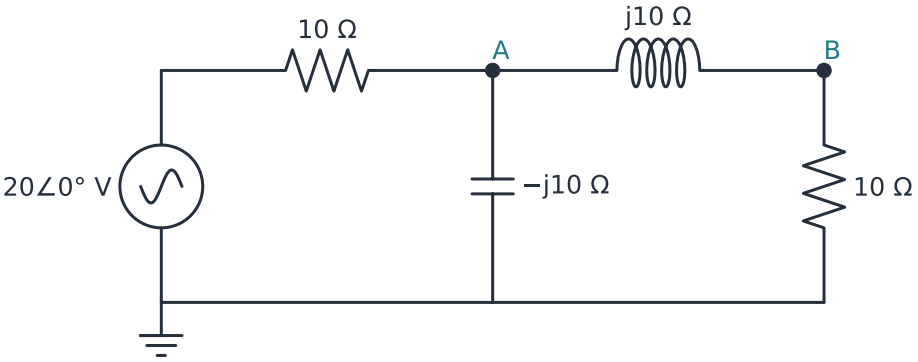


Fig 1.15 AC node circuit for Example 1.11.

SOLVED EXAMPLE 1.11 · AC NODE

Find the node voltages V_A and V_B in Fig 1.15.

- Node A: $(V_A - 20)/10 + V_A/(-j10) + (V_A - V_B)/(j10) = 0$. Since $1/(-j10) = j0.1$ and $1/(j10) = -j0.1$, the V_A terms give $0.1 + j0.1 - j0.1 = 0.1$. So $0.1V_A + j0.1V_B = 2$.
- Node B: $(V_B - V_A)/(j10) + V_B/10 = 0 \Rightarrow j0.1V_A + (0.1 - j0.1)V_B = 0 \Rightarrow V_A = (1 + j)V_B$.
- Substitute: $0.1(1 + j)V_B + j0.1V_B = (0.1 + j0.2)V_B = 2 \Rightarrow V_B = 2/(0.1 + j0.2) = 4 - j8$ V.
- $V_A = (1 + j)(4 - j8) = 4 - j8 + j4 + 8 = 12 - j4$ V.

$V_A = 12 - j4 = 12.65\angle -18.43°$ V, $V_B = 4 - j8 = 8.94\angle -63.43°$ V

COMMON MISTAKE

Adding magnitudes of phasors ($|I_1| - |I_2| = 0$ here, yet $I_C = 3.16$ A!). Always subtract in rectangular form, then convert to polar.

INTERVIEW CORNER

Q. Why do we use phasors?

A. In a linear circuit driven at one frequency, every voltage and current is a sinusoid of that same frequency; only amplitude and phase differ. Phasors keep just those two numbers and turn differential equations into algebra ($d/dt \rightarrow j\omega$).

Q. Is the impedance of a capacitor at DC zero or infinite?

A. Infinite ($\omega = 0 \Rightarrow 1/\omega C \rightarrow \infty$) — an open circuit. An inductor at DC is a short. At very high frequency it is the reverse.

✓ Module 1 — Quick revision

Network vs circuit: circuit = network with at least one closed path.

Passive sign convention: current into + ⇒ p = vi absorbed.

Element laws: v = Ri, v = L di/dt, i = C dv/dt. L current and C voltage cannot jump.

Practical sources: V source + series R_s; I source ‖ R_p. Ideal: $R_s = 0$, $R_p = \infty$.

Dependent sources: VCVS μ, CCVS r (Ω), VCCS g (S), CCCS β. Never kill them.

Source transformation: V_s series R ⇔ $I_s = V_s/R$ parallel R. Not for ideal sources.

KCL: Σ i at a node = 0 (charge). **KVL:** Σ v round a loop = 0 (energy).

Mesh: B − N + 1 equations; [R][I] = [V]; planar only.

Node: N − 1 equations; [G][V] = [I]; works for non-planar too.

Supermesh: current source between two meshes → 1 KVL round both + $I_s = I_a - I_b$.

Supernode: voltage source between two non-ref nodes → 1 KCL for bubble + $V_a - V_b = V_s$.

Series with ideal I-source: a resistor there does not affect any current.

AC: R, jωL, 1/(jωC); same methods with complex numbers; P = Re(VI*).

Dependent sources make [R]/[G] unsymmetric.

Practice — short questions (3 marks)

1. Distinguish between a network and a circuit with an example.
2. Draw the four types of dependent sources and give the unit of each controlling constant.
3. Explain the difference between ideal and practical voltage sources with V-I characteristics.
4. State and explain KCL and KVL. On which conservation laws are they based?
5. Convert a 20 V source with 4 Ω series resistance into an equivalent current source. [5 A ‖ 4 Ω]
6. What is a supermesh? When is it used?
7. What is a supernode? Write the equations needed for it.
8. A network has 8 branches and 5 nodes. How many mesh equations and node equations are needed? [4 and 4]

Practice — long questions (9/14 marks)

1. Using mesh analysis, find the current in the 6 Ω resistor of Fig 1.7 if the 5 V source is reversed.
   [Ans: $I_1 = I_2 = 5$ A ⇒ zero current in 6 Ω — interpret why]
2. In Fig 1.10 replace the 12 V source by 24 V and find V_1, V_2 by node analysis. [Ans: $V_1 = 124/19 \approx 6.53$ V, $V_2 = 296/19 \approx 15.58$ V]

3. Explain supermesh analysis with an example and solve Fig 1.12 with the 16 V source changed to 26 V. [Ans: I_1 = 4 A, I_2 = 6 A]
4. Using source transformation, reduce Fig 1.4 to a single loop and find the power in the 4 Ω resistor. [Ans: 9 W]
5. In Fig 1.11 change the VCCS to $0.25V_1$ and find V_1, V_2. [Ans: 36/7 ≈ 5.14 V, 40/7 ≈ 5.71 V]
6. Solve Fig 1.14 by node analysis (take the capacitor top as the only unknown node) and confirm $I_C = 3.162\angle 71.57°$ A.

02

Network Theorems

> *Theorems are shortcuts: they shrink a big circuit into a small one. Thevenin with dependent sources and maximum power transfer come up in almost every core-electronics interview.*

SYLLABUS CHECKLIST

- ☐ Superposition theorem
- ☐ Reciprocity theorem
- ☐ Thevenin's theorem
- ☐ Norton's theorem
- ☐ Millman's theorem
- ☐ Maximum power transfer theorem
- ☐ All applied to DC and AC networks with independent and dependent sources

11 hours · CO2 · Apply network theorems to DC and AC networks

Module 1 gave you a universal method: write mesh or node equations and solve. It always works — but it can be slow, and it hides the *meaning*. The theorems in this module answer questions like "what does the rest of the circuit look like from these two terminals?" or "how much does each source contribute?". They all rest on one property: the circuit is **linear**.

KEY IDEA — LINEARITY

A circuit made of R, L, C, independent sources and *linear* dependent sources obeys two rules: **scaling** (double every source → every voltage and current doubles) and **additivity** (the response to two sources together is the sum of responses to each alone). Superposition, Thevenin and Norton are just consequences of these two rules.

2.1 Superposition theorem

Statement. In a linear network with several independent sources, the voltage across (or current through) any element equals the algebraic sum of the voltages (or currents) produced by each independent source acting *alone*, with all other independent sources turned off.

THINK OF IT LIKE

Two people pushing a box. The total movement is what person A alone would cause plus what person B alone would cause — as long as the floor behaves the same way no matter how hard they push (that is linearity).

How to "turn off" a source

Source	Turn it off by…	Why
Independent voltage source	replacing it with a **short circuit**	a 0 V source is a wire
Independent current source	replacing it with an **open circuit**	a 0 A source is a break
Dependent source	**never** turn it off — leave it in every sub-circuit	its value depends on the circuit itself
Practical source	kill only the ideal part, keep its internal resistance	R_s is a real resistor

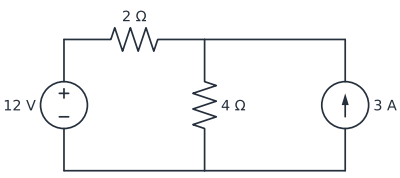


Fig 2.1 Original circuit.

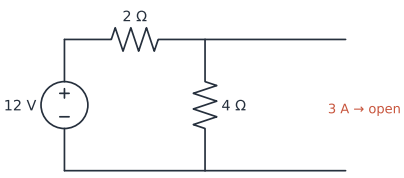


Fig 2.2 12 V acting alone.

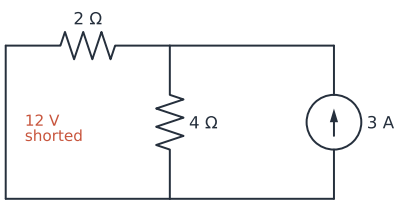


Fig 2.3 3 A acting alone.

SOLVED EXAMPLE 2.1 · KTU STYLE

Using superposition, find the current I in the 4 Ω resistor of Fig 2.1. Also check whether the power in the 4 Ω can be found by superposition.

- 12 V alone (3 A → open), Fig 2.2: series loop 2 + 4 = 6 Ω ⇒ $I' = 12/6 = 2$ A (downward).
- 3 A alone (12 V → short), Fig 2.3: 3 A splits between 2 Ω and 4 Ω. Current divider: $I'' = 3 \times 2/(2 + 4) = 1$ A (downward).
- Total: $I = I' + I'' = 3$ A downward.
- Power: true $P = 3^2 \times 4 = 36$ W. "Superposed" power: $2^2 \times 4 + 1^2 \times 4 = 16 + 4 = 20$ W ✗ — wrong, because power is proportional to I^2, which is not linear.

I = 3 A; power cannot be superposed (36 W, not 20 W)

COMMON MISTAKE

Superposing **power**. $(I' + I'')^2 \neq I'^2 + I''^2$. Find the total current or voltage first, then compute power. (Exception: sources at *different frequencies* — their powers do add, see Example 2.11.)

Superposition with a dependent source

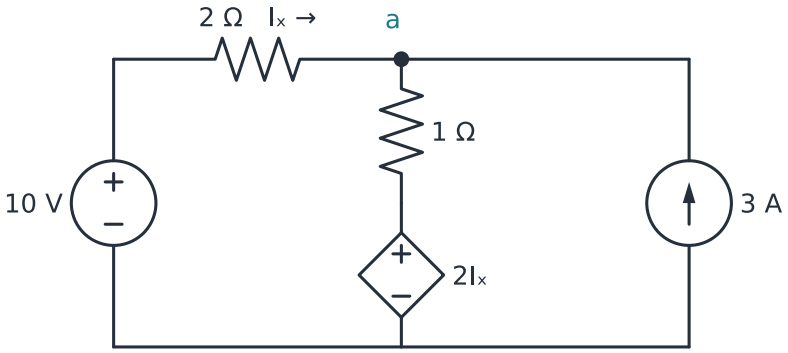


Fig 2.4 Circuit for Example 2.2.

SOLVED EXAMPLE 2.2 · DEPENDENT SOURCE

Find I_x in Fig 2.4 using superposition. The CCVS ($2I_x$) is in series with 1 Ω in the middle branch.

- Node a voltage: $V_a = 1 \times$ (current down the middle) + $2I_x$.
- **10 V alone** (3 A open): middle current = $I_x' \Rightarrow V_a = I_x' + 2I_x' = 3I_x'$. Also $I_x' = (10 - 3I_x')/2 \Rightarrow 5I_x' = 10 \Rightarrow I_x' = 2$ A.
- **3 A alone** (10 V shorted): middle current = $I_x'' + 3 \Rightarrow V_a = 3I_x'' + 3$. The 2 Ω now goes to ground, so $I_x'' = (0 - V_a)/2 \Rightarrow V_a = -2I_x''$. Hence $-2I_x'' = 3I_x'' + 3 \Rightarrow I_x'' = -0.6$ A.
- $I_x = 2 + (-0.6) = 1.4$ A. (Check by direct node analysis: $V_a = 7.2$ V, $I_x = (10 - 7.2)/2 = 1.4$ A √.)

$I_x = 1.4$ A

INTERVIEW CORNER

Q. Is superposition valid for a circuit with a diode?

A. No — a diode is non-linear. Superposition needs linearity. (We do use it for small-signal analysis after linearising around a bias point.)

Q. Can superposition be applied to dependent sources one at a time?

A. No. Only independent sources are switched on and off. Dependent sources remain in every sub-circuit.

2.2 Thevenin's theorem

Suppose you only care about what happens at two terminals a-b — maybe you will connect different loads there. Thevenin's theorem says: **however complicated the linear network behind a-b, it behaves exactly like one voltage source V_{th} in series with one resistance R_{th}.**

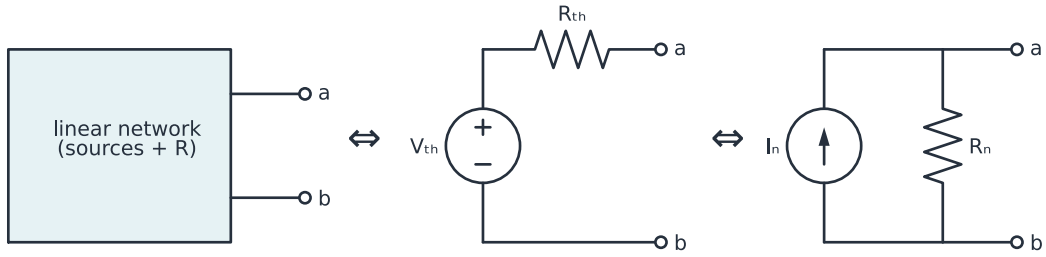


Fig 2.5 Any linear two-terminal network ⇔ Thevenin equivalent ⇔ Norton equivalent.

THINK OF IT LIKE

Your phone charger box has transformers, diodes and ICs inside, but your phone only "sees" something like a 5 V source with a small internal resistance. Thevenin is the art of describing a black box by what it looks like from outside.

Finding V_{th} and R_{th}

1. Remove the load. **V_{th} = open-circuit voltage** V_{oc} across a-b.
2. **R_{th}**, by one of three methods:
   - **Method A (independent sources only):** kill all independent sources and find the equivalent resistance seen from a-b.
   - **Method B (any circuit):** short a-b, find I_{sc}; then $R_{th} = V_{oc}/I_{sc}$.
   - **Method C (any circuit, best when there are no independent sources):** kill independent sources, apply a test source (1 A or 1 V) at a-b, $R_{th} = V_{test}/I_{test}$.
3. Draw V_{th} in series with R_{th}, reconnect the load: $I_L = V_{th}/(R_{th} + R_L)$.

Why it works (proof idea). Drive terminals a-b with a current source I. By superposition, V_{ab} = (contribution of internal sources with I = 0) + (contribution of I with internal sources killed) = $V_{oc} - R_{th} \cdot I$. That straight-line V-I relation is exactly what a source V_{oc} in series with R_{th} gives.

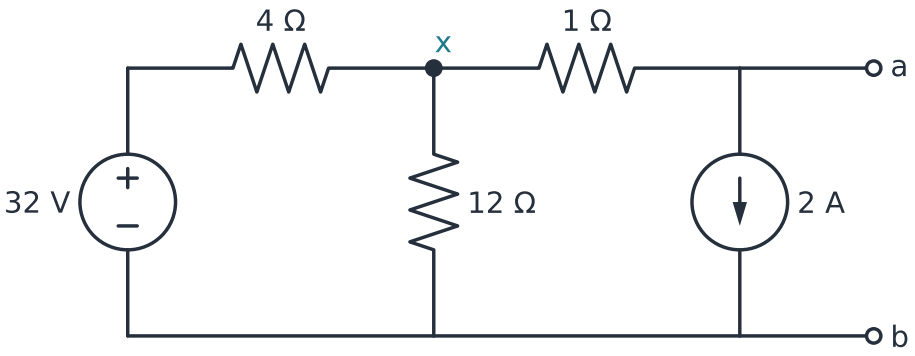


Fig 2.6 Circuit for Examples 2.3 and 2.4.

SOLVED EXAMPLE 2.3 · KTU STYLE

Find the Thevenin equivalent at a-b of Fig 2.6 (the 2 A source draws current from a to b). Then find the load current for R_L = 2 Ω, 4 Ω and 12 Ω.

- V_{oc}: with a-b open, all 2 A flows through the 1 Ω. KCL at x: $(V_x - 32)/4 + V_x/12 + 2 = 0 \Rightarrow V_x(1/3) = 6 \Rightarrow V_x = 18$ V.
- $V_{th} = V_x - 2 \times 1 = 16$ V.
- R_{th}: short 32 V, open 2 A ⇒ $(4 \parallel 12) + 1 = 3 + 1 = 4$ Ω.
- $I_L = 16/(4 + R_L)$: R_L = 2 Ω → 2.667 A (14.22 W); 4 Ω → 2 A (16 W); 12 Ω → 1 A (12 W).

V_{th} = 16 V, R_{th} = 4 Ω; I_L = 2.667 A, 2 A, 1 A

See how the load power peaked at $R_L = 4\ \Omega = R_{th}$? Hold that thought for Section 2.5.

2.3 Norton's theorem

Norton's theorem is Thevenin's twin: the same network looks like a **current source I_N in parallel with R_N**. The two are linked by source transformation.

THEVENIN ⇔ NORTON

$$I_N = I_{sc} \text{ (short-circuit current at a-b)} \qquad R_N = R_{th} \qquad V_{th} = I_N \cdot R_N$$

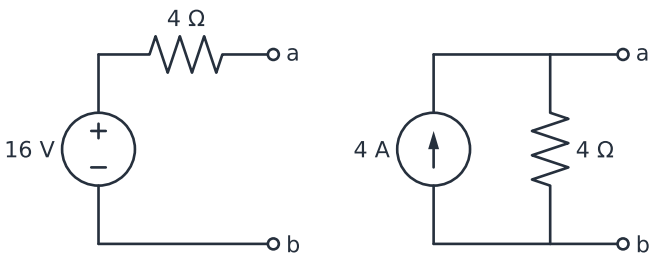


Fig 2.7 Thevenin and Norton equivalents of Fig 2.6.

SOLVED EXAMPLE 2.4 · NORTON

Find the Norton equivalent of Fig 2.6 directly from the short-circuit current, and confirm it matches Example 2.3.

- Short a-b ($V_a = 0$). KCL at x: $(V_x - 32)/4 + V_x/12 + V_x/1 = 0 \Rightarrow V_x(4/3) = 8 \Rightarrow V_x = 6$ V.
- Current through 1 Ω towards a = 6 A. Of this, 2 A goes into the current source; the short carries $I_{sc} = 6 - 2 = 4$ A.
- $R_N = R_{th} = 4\ \Omega$. Check: $I_N R_N = 16\ \text{V} = V_{th}$ √.

$I_N = 4$ A, $R_N = 4\ \Omega$ (Fig 2.7)

EXAM TIP

Find any **two** of V_{oc}, I_{sc}, R_{th} and the third follows. Use the third as a free cross-check — examiners love answers that are verified.

2.4 Thevenin and Norton with dependent sources Frequently asked

This is the single most popular theorem question in interviews and GATE. The rule is simple: **you cannot find R_{th} by just "looking into" the circuit**, because the dependent source changes its value when you connect something. Use Method B (V_{oc}/I_{sc}) or Method C (test source).

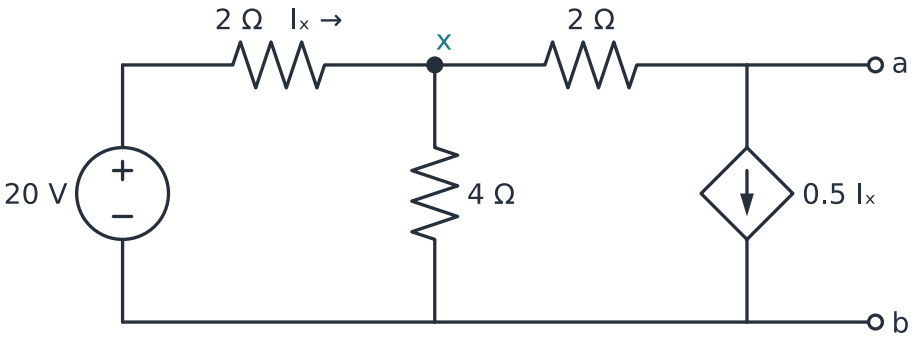


Fig 2.8 Circuit with a CCCS for Example 2.5. I_x is the current in the first 2 Ω.

SOLVED EXAMPLE 2.5 · PLACEMENT FAVOURITE

Find the Thevenin equivalent at a-b in Fig 2.8 and the maximum power that can be delivered to a resistive load.

- **Open circuit.** The only path out of a is the CCCS, so the second 2 Ω carries $0.5I_x$. KCL at x: $I_x = V_x/4 + 0.5I_x \Rightarrow I_x = V_x/2$. Also $I_x = (20 - V_x)/2$. Equating: $V_x = 10$ V, $I_x = 5$ A.
- $V_{oc} = V_x - 2(0.5I_x) = 10 - 5 = 5$ V.
- **Short circuit** ($V_a = 0$). KCL at x: $I_x = V_x/4 + V_x/2 = 0.75V_x$; with $I_x = (20 - V_x)/2 \Rightarrow V_x = 8$ V, $I_x = 6$ A.
- At a: current arriving = $V_x/2 = 4$ A; CCCS takes $0.5 \times 6 = 3$ A $\Rightarrow I_{sc} = 1$ A.
- $R_{th} = V_{oc}/I_{sc} = 5$ Ω. (Test-source check: killing 20 V and injecting 1 A at a gives $V_a = 5$ V ✓.)
- $P_{max} = V_{th}^2/(4R_{th}) = 25/20 = 1.25$ W at $R_L = 5$ Ω.

$V_{th} = 5$ V, $R_{th} = 5$ Ω, $I_N = 1$ A, $P_{max} = 1.25$ W

Notice: if you had wrongly "killed" the CCCS, you would get $R_{th} = 2 + (2 \parallel 4) = 3.33$ Ω — wrong.

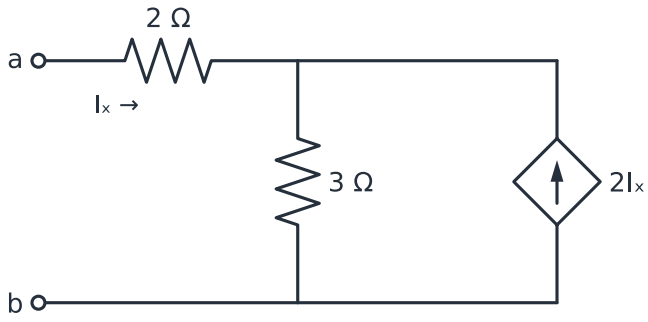


Fig 2.9 A network with only a dependent source (Example 2.6).

SOLVED EXAMPLE 2.6 · TEST SOURCE

Find the Thevenin equivalent of Fig 2.9 at a-b. The CCCS pushes $2I_x$ upward into the middle node.

- No independent source $\Rightarrow V_{th} = 0$ (a network cannot produce voltage on its own).
- Apply a test current of 1 A into a: $I_x = 1$ A. At the middle node: current into the 3 Ω = $I_x + 2I_x = 3$ A $\Rightarrow V_n = 9$ V.
- $V_a = 2 \times 1 + 9 = 11$ V $\Rightarrow R_{th} = 11/1 = 11$ Ω.
- Bonus: if the CCCS pointed downward, the 3 Ω current would be $I_x - 2I_x = -1$ A $\Rightarrow V_a = 2 - 3 = -1$ V $\Rightarrow R_{th} = -1$ Ω. Negative resistance is possible with dependent sources — that is how oscillators work.

$V_{th} = 0$, $R_{th} = 11$ Ω (would be −1 Ω with the source reversed)

COMMON MISTAKE

Using V_{oc}/I_{sc} when there is no independent source (0/0!). Use a test source instead.

INTERVIEW CORNER

Q. How would you measure the Thevenin resistance of a real circuit in the lab?
A. Measure V_{oc} with a high-impedance voltmeter, then connect a known load R_L and measure V_L: $R_{th} = R_L(V_{oc}/V_L - 1)$. Shorting the output to measure I_{sc} works too but may damage the source.

Q. What is the Thevenin resistance looking into an ideal op-amp voltage follower's output?
A. Zero — the output is an ideal voltage source. The finite open-loop gain A reduces a real op-amp's output resistance R_o to roughly $R_o/(1 + A)$.

Q. Can R_{th} be negative?
A. Yes, only with dependent (active) sources. A passive network always has $R_{th} \geq 0$.

2.5 Maximum power transfer theorem

Connect a load R_L to a source with internal resistance R_{th}. If R_L is tiny, lots of current flows but almost no voltage appears across the load. If R_L is huge, there is lots of voltage but hardly any current. Somewhere in between, the power V·I is largest.

Derivation. With the Thevenin equivalent, $P_L = I^2 R_L = \dfrac{V_{th}^2 R_L}{(R_{th} + R_L)^2}$. Set $dP_L/dR_L = 0$:

DERIVATION

$$\frac{dP_L}{dR_L} = V_{th}^2 \frac{(R_{th} + R_L)^2 - 2R_L(R_{th} + R_L)}{(R_{th} + R_L)^4} = 0 \Rightarrow R_{th} + R_L = 2R_L \Rightarrow \mathbf{R_L = R_{th}}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{I_N^2 R_N}{4} \qquad \text{efficiency at } P_{max}\text{: } \eta = 50\ \%$$

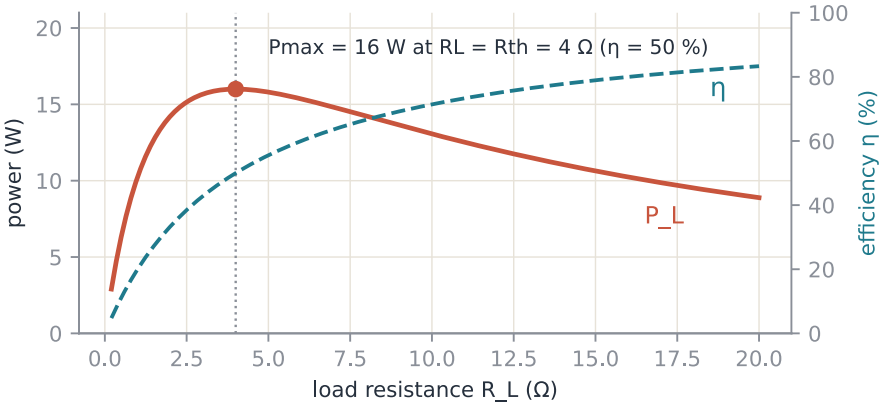


Fig 2.10 Load power and efficiency vs R_L for V_{th} = 16 V, R_{th} = 4 Ω (Example 2.3's circuit).

AC version

Load allowed	Condition for maximum power	P_{max} (rms V_{th})
Any complex $Z_L = R_L + jX_L$	$Z_L = Z_{th}^*$ (conjugate match: $R_L = R_{th}$, $X_L = -X_{th}$)	$\lvert V_{th}\rvert^2/(4R_{th})$
Pure resistance R_L only	$R_L = \lvert Z_{th}\rvert = \sqrt{(R_{th}^2 + X_{th}^2)}$	$\lvert I\rvert^2 R_L$
R_L variable, X_L fixed	$R_L = \sqrt{(R_{th}^2 + (X_{th} + X_L)^2)}$	$\lvert I\rvert^2 R_L$

SOLVED EXAMPLE 2.7 · AC MAXIMUM POWER

A source has V_{th} = 10 V (rms) and Z_{th} = 3 + j4 Ω. Find the maximum load power if (a) Z_L can be any impedance, (b) the load must be a pure resistor.

▸ (a) $Z_L = 3 - j4$ Ω. Reactances cancel; total impedance 6 Ω ⇒ I = 10/6 A ⇒ P = $(10/6)^2 \times 3$ = 100/12 = 8.33 W.

▸ (b) $R_L = |3 + j4| = 5$ Ω. I = 10/|8 + j4| = 10/8.944 = 1.118 A ⇒ P = $1.118^2 \times 5$ = 6.25 W.

(a) 8.33 W with $Z_L = 3 - j4$ Ω; (b) 6.25 W with R_L = 5 Ω

INTERVIEW CORNER

Q. If maximum power transfer gives only 50 % efficiency, why don't power utilities use it?

A. Utilities want high *efficiency*, so they make $R_L \gg R_{source}$. MPT matters where the signal is weak and every microwatt counts — antennas, RF front-ends, audio amplifier-to-speaker, sensors.

Q. Why are RF systems matched to 50 Ω?

A. Matching source, line and load impedances gives maximum power transfer and no reflections on the transmission line. 50 Ω is a historical compromise between low loss (≈77 Ω for air coax) and high power handling (≈30 Ω).

Q. The open-circuit voltage of a source is 12 V and its short-circuit current is 3 A. What load draws maximum power, and how much?

A. R_{th} = 12/3 = 4 Ω; load 4 Ω; $P_{max} = 12^2/16$ = 9 W.

2.6 Reciprocity theorem

Statement. In a linear, bilateral network with a **single** independent source and no dependent sources, if a voltage source E in branch 1 produces a current I in branch 2, then the same source moved to branch 2 produces the same current I in branch 1. In short: the ratio **excitation/response stays the same when their positions are swapped**.

THINK OF IT LIKE

Two people in a big hall full of pillars. If A's shout reaches B at some loudness, B's identical shout reaches A at the same loudness — the hall (passive, bilateral) treats both directions alike.

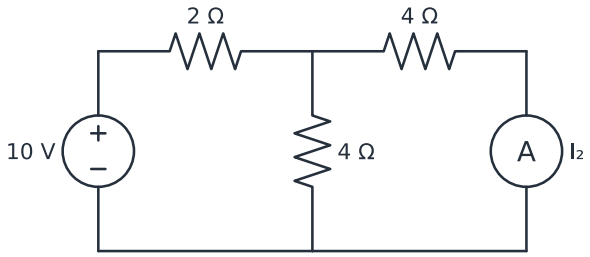


Fig 2.11 Source at left, ammeter at right.

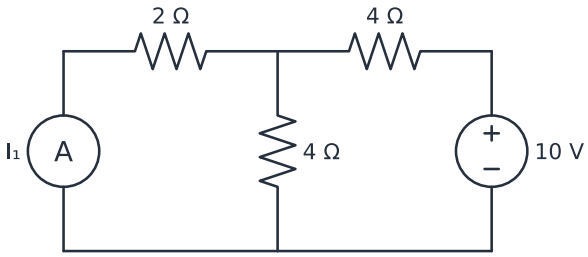


Fig 2.12 Positions swapped.

SOLVED EXAMPLE 2.8 · VERIFY RECIPROCITY

Verify the reciprocity theorem for the network of Figs 2.11 and 2.12 (ammeters ideal, 0 Ω).

- Fig 2.11: resistance seen by source = 2 + (4 ∥ 4) = 4 Ω ⇒ I_s = 10/4 = 2.5 A. Current divider: I_2 = 2.5 × 4/(4 + 4) = 1.25 A.
- Fig 2.12: resistance seen by source = 4 + (4 ∥ 2) = 4 + 4/3 = 16/3 Ω ⇒ I_s = 1.875 A. I_1 = 1.875 × 4/(4 + 2) = 1.25 A.
- E/I is 10/1.25 = 8 Ω in both cases ⇒ reciprocity verified. (This 8 Ω is the "transfer resistance".)

$I_2 = I_1 = 1.25$ A ✓

COMMON MISTAKE

Applying reciprocity to circuits with dependent sources (transistor models) — they are generally **not** reciprocal. Also, when swapping, the ammeter replaces the source and the source replaces the ammeter; the rest of the circuit must stay untouched.

INTERVIEW CORNER

Q. Which theorems need linearity and which need bilateral elements?

A. Superposition, Thevenin, Norton, Millman need linearity. Reciprocity needs linear *and* bilateral elements with no dependent sources. MPT itself just needs a Thevenin equivalent, so linearity of the source side.

2.7 Millman's theorem

Millman's theorem is a shortcut for many voltage sources (each with its own series resistance) connected **in parallel** between the same two nodes. They combine into one equivalent source.

MILLMAN'S THEOREM

$$V_{eq} = \frac{E_1G_1 + E_2G_2 + \ldots + E_nG_n}{G_1 + G_2 + \ldots + G_n} \qquad R_{eq} = \frac{1}{G_1 + G_2 + \ldots + G_n} \qquad (G_k = 1/R_k)$$

AC: replace G by admittance Y = 1/Z.

Why it works. Transform each branch into a current source E_kG_k in parallel with G_k. Parallel current sources add; parallel conductances add. Transform back: $V_{eq} = (\Sigma\ E_kG_k)/(\Sigma\ G_k)$. It is really just one nodal equation in disguise.

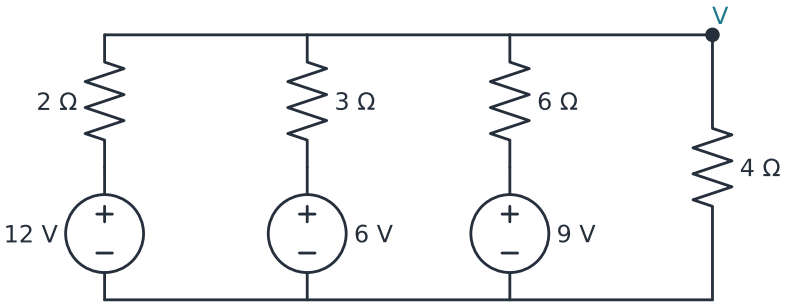


Fig 2.13 Three parallel sources feeding a 4 Ω load (Example 2.9).

SOLVED EXAMPLE 2.9 · MILLMAN

Using Millman's theorem, find the current and voltage across the 4 Ω load in Fig 2.13.

- $G = 1/2 + 1/3 + 1/6 = 1\ S \Rightarrow R_{eq} = 1\ \Omega$.
- $\Sigma\ EG = 12/2 + 6/3 + 9/6 = 6 + 2 + 1.5 = 9.5\ A \Rightarrow V_{eq} = 9.5/1 = 9.5\ V$.
- $I_L = 9.5/(1 + 4) = 1.9\ A$; $V = 1.9 \times 4 = 7.6\ V$.

$I_L = 1.9$ A, $V_L = 7.6$ V

EXAM TIP

Watch polarity: if a source is connected upside down, use −E for it in the numerator. A branch with only a resistor (no source) still adds its G to the denominator, with E = 0.

2.8 Theorems in AC networks

All the theorems work for sinusoidal steady state exactly as for DC — just use impedances and phasors. A few things to keep in mind:

- Thevenin/Norton: V_{th} and I_N are phasors; $Z_{th} = V_{oc}/I_{sc}$ is complex.
- Superposition with sources at **different frequencies**: solve each source separately (each with its own jω), add the **time-domain** answers — never add phasors of different frequencies.
- MPT: conjugate matching (Section 2.5).

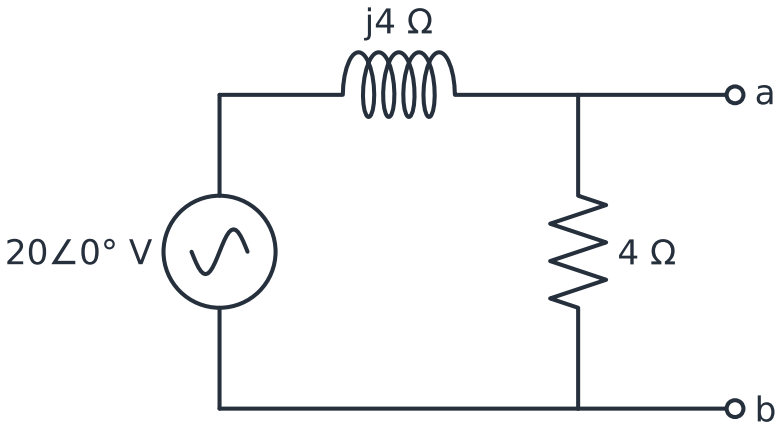


Fig 2.14 AC circuit for Example 2.10.

SOLVED EXAMPLE 2.10 · AC THEVENIN AND NORTON

For Fig 2.14 (rms values) find the Thevenin and Norton equivalents at a-b, the current in a 2 Ω load, and the load for maximum power.

- $V_{th} = 20 \times \frac{4}{4 + j4} = \frac{80}{4 + j4} = 10 - j10 = 14.14\angle -45°$ V.
- $Z_{th} = j4 \parallel 4 = \frac{j16}{4 + j4} = 2 + j2\ \Omega$.
- $I_N = V_{th}/Z_{th} = (10 - j10)/(2 + j2) = -j5 = 5\angle -90°$ A. (Check: shorting a-b, $I_{sc} = 20/j4 = -j5$ √.)
- 2 Ω load: $I_L = (10 - j10)/(4 + j2) = 1 - j3 = 3.162\angle -71.57°$ A; $P = 3.162^2 \times 2 = 20$ W.
- Max power: $Z_L = 2 - j2\ \Omega \Rightarrow P_{max} = |V_{th}|^2/(4 \times 2) = 200/8 = 25$ W.

$V_{th} = 14.14\angle -45°$ V, $Z_{th} = 2 + j2\ \Omega$, $I_N = 5\angle -90°$ A; $I_L = 3.162\angle -71.57°$ A; $P_{max} = 25$ W

SOLVED EXAMPLE 2.11 · TWO FREQUENCIES (GATE STYLE)

A source v(t) = 10 + 10 cos 1000t V drives a series R = 10 Ω, L = 10 mH. Find i(t) and the average power in R.

- DC part (ω = 0, L is a short): $I_0 = 10/10 = 1$ A.
- AC part: $Z = 10 + j(1000)(0.01) = 10 + j10 = 14.14\angle 45°\ \Omega \Rightarrow I_m = 10/14.14\angle 45° = 0.707\angle -45°$ A (peak).
- $i(t) = 1 + 0.707 \cos(1000t - 45°)$ A.
- Different frequencies ⇒ powers add: $P = 1^2 \times 10 + (0.707^2/2) \times 10 = 10 + 2.5 = 12.5$ W.

$i(t) = 1 + 0.707 \cos(1000t - 45°)$ A, P = 12.5 W

✓ Module 2 — Quick revision

Superposition: one independent source at a time; V-source → short, I-source → open; dependent sources stay. Not for power.

Thevenin: $V_{th} = V_{oc}$; R_{th} by killing sources, V_{oc}/I_{sc}, or test source.

Norton: $I_N = I_{sc}$, $R_N = R_{th}$, $V_{th} = I_N R_N$.

Dependent sources: never kill them; use V_{oc}/I_{sc} or a test source; R_{th} can be negative.

Only dependent sources: $V_{th} = 0$, find R_{th} with a test source.

MPT (DC): $R_L = R_{th}$, $P_{max} = V_{th}^2/4R_{th}$, $\eta = 50\ \%$.

MPT (AC): $Z_L = Z_{th}^*$; resistive-only load: $R_L = |Z_{th}|$.

Reciprocity: single source, linear bilateral, no dependent sources; E/I unchanged on swapping.

Millman: $V = \Sigma E_k G_k / \Sigma G_k$, $R = 1/\Sigma G_k$.

Different frequencies: solve separately, add in time domain; powers add.

Practice — short questions (3 marks)

1. State the superposition theorem. Why can't it be used to find power?
2. State Thevenin's theorem and list the steps to find the Thevenin equivalent.
3. How is R_{th} found when the network contains dependent sources?
4. State and prove the maximum power transfer theorem for a DC network.
5. State the reciprocity theorem. Give two conditions for its validity.
6. State Millman's theorem. Find V_{eq} for 10 V/5 Ω and 20 V/5 Ω in parallel. [15 V, 2.5 Ω]
7. For $Z_{th} = 6 + j8\ \Omega$, what pure resistance absorbs maximum power? [10 Ω]
8. A source has $V_{oc} = 24$ V and $I_{sc} = 4$ A. Find P_{max}. [24 W]

Practice — long questions (9/14 marks)

1. In Fig 2.1, change the source to 24 V and find the 4 Ω current by superposition. [Ans: 4 + 1 = 5 A]
2. Find the Thevenin and Norton equivalents of Fig 2.6 if the 2 A source is reversed. [Ans: $V_{th} = 32$ V, $R_{th} = 4\ \Omega$, $I_N = 8$ A]
3. Find R_{th} for Fig 2.8 if the CCCS is changed to $1 \cdot I_x$. Comment on the result. [Ans: $V_{oc} = -20$ V, $I_{sc} = -2$ A, $R_{th} = 10\ \Omega$ — the polarity of V_{th} flips!]
4. Derive the condition for maximum power transfer to a load $Z_L = R_L + jX_L$ when both R_L and X_L are variable.
5. Verify reciprocity for Fig 2.11 with the middle 4 Ω changed to 2 Ω. [Ans: I = 1.0 A both ways]
6. Using Thevenin's theorem, find the current in the inductor branch of Fig 1.14 (Module 1). [Ans: $2.236\angle -63.43°$ A]

03

Laplace Transforms & Transient Analysis

> *What happens in the first few milliseconds after a switch flips? Laplace turns those messy differential equations into simple algebra — and "explain the time constant" is a standard interview opener.*

SYLLABUS CHECKLIST

- ☐ Review of Laplace transform: definition, properties and theorems
- ☐ Laplace transforms of standard signals
- ☐ Inverse Laplace transform by partial fractions
- ☐ s-domain representation of R, L and C, including initial conditions
- ☐ Transient analysis of RL, RC and RLC circuits with DC, impulse, step and sinusoidal inputs
- ☐ Low-pass and high-pass RC circuits: response to step and pulse, time constant

11 hours · CO3 · Analyse the transient behaviour of networks using Laplace transforms

3.1 Why transients, and why Laplace?

In Modules 1 and 2 every circuit had settled down: DC values were constant, AC phasors were steady. But real circuits get **switched** — a power supply turns on, a relay closes, a digital gate toggles. Inductors and capacitors store energy, and stored energy cannot change in zero time. So for a short while after switching, the circuit is "in between" its old and new states. That in-between behaviour is the **transient**.

> **THINK OF IT LIKE**
>
> Pushing a child on a swing. A resistor is like sand under the swing — it wastes energy. An inductor is the child's *momentum* (current can't stop instantly). A capacitor is the *height* of the swing (voltage can't jump instantly). The transient is how the swing settles after your push.

The two rules that decide everything at t = 0

Element	Cannot change instantly	At $t = 0^+$ (just after switching), if uncharged	At $t \to \infty$ with DC
Inductor L	current i_L	open circuit ($i_L = 0$); if carrying I_0, a current source I_0	short circuit
Capacitor C	voltage v_C	short circuit ($v_C = 0$); if charged to V_0, a voltage source V_0	open circuit
Resistor R	— (no memory)	R	R

Why Laplace? KVL on an RL or RLC circuit gives a *differential* equation. Solving it classically means guessing a solution, finding constants from initial conditions, and repeating for every input. The Laplace transform changes d/dt into multiplication by s, so the differential equation becomes an algebraic one — and the initial conditions get built in automatically.

3.2 Laplace transforms of standard signals

The (one-sided) Laplace transform of a signal f(t) defined for $t \geq 0$ is

> **DEFINITION**
>
> $$F(s) = \mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt, \quad s = \sigma + j\omega \text{ (complex frequency)}$$

Think of e^{-st} as a family of "test signals" — decaying, growing or oscillating exponentials. F(s) tells you how much of each one is inside f(t). Starting the integral at 0^- lets impulses at t = 0 and initial conditions be included. The integral converges only for Re(s) larger than some σ_0 — the **region of convergence**; for the causal signals in circuits it always exists, so we rarely worry about it.

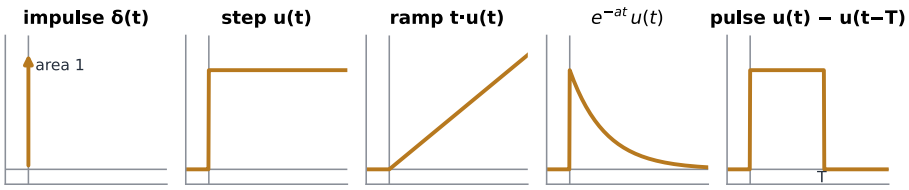


Fig 3.1 The standard test signals used in transient analysis.

Three quick derivations

- **Unit step:** $\mathscr{L}\{u(t)\} = \int_0^\infty e^{-st}\,dt = [-e^{-st}/s]_0^\infty = 1/s$.
- **Exponential:** $\mathscr{L}\{e^{-at}\} = \int_0^\infty e^{-(s+a)t}\,dt = 1/(s+a)$.
- **Sine:** $\sin \omega t = (e^{j\omega t} - e^{-j\omega t})/2j \Rightarrow \mathscr{L} = \frac{1}{2j}\left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega}\right] = \frac{\omega}{s^2+\omega^2}$.

f(t), t ≥ 0	F(s)	f(t), t ≥ 0	F(s)
δ(t) (unit impulse)	1	sin ωt	$\omega/(s^2+\omega^2)$
u(t) (unit step)	1/s	cos ωt	$s/(s^2+\omega^2)$
t (ramp)	$1/s^2$	e^{-at} sin ωt	$\omega/[(s+a)^2+\omega^2]$
t^n	$n!/s^{n+1}$	e^{-at} cos ωt	$(s+a)/[(s+a)^2+\omega^2]$
e^{-at}	$1/(s+a)$	sinh at	$a/(s^2-a^2)$
$t\,e^{-at}$	$1/(s+a)^2$	cosh at	$s/(s^2-a^2)$

3.3 Laplace transform theorems (properties)

You will rarely integrate from scratch. Instead, build transforms from the table using these properties ($F(s) = \mathscr{L}\{f(t)\}$):

Property	Time domain	s-domain
Linearity	$a\,f_1(t) + b\,f_2(t)$	$a\,F_1(s) + b\,F_2(s)$
Time shift (delay)	$f(t-T)\,u(t-T)$	$e^{-sT}\,F(s)$
Frequency shift	$e^{-at}\,f(t)$	$F(s+a)$
Time scaling	$f(at)$, $a>0$	$(1/a)\,F(s/a)$
Differentiation	df/dt | d^2f/dt^2	$sF(s) - f(0^-)$ | $s^2F(s) - s\,f(0^-) - f'(0^-)$
Integration	$\int_0^t f(\tau)\,d\tau$	$F(s)/s$
Multiplication by t	$t\,f(t)$	$-dF(s)/ds$
Convolution	$f_1(t) * f_2(t)$	$F_1(s)\,F_2(s)$
Initial value theorem	$f(0^+)$	$\lim_{s\to\infty} sF(s)$
Final value theorem	$f(\infty)$	$\lim_{s\to 0} sF(s)$ (only if all poles of sF(s) are in the left half-plane)

Proof of the differentiation rule (KTU asks this). Integrate by parts with $u = e^{-st}$, $dv = f'(t)dt$: $\mathcal{L}\{f'\} = [f(t)e^{-st}]_{0^-}^{\infty} + s\int f(t)e^{-st}dt = -f(0^-) + sF(s)$. That little "$-f(0^-)$" is exactly how initial inductor currents and capacitor voltages enter our circuit equations.

SOLVED EXAMPLE 3.1 · TRANSFORMS

Find the Laplace transforms of (a) $f(t) = 2u(t) - 3e^{-2t} + 4\cos 3t$, (b) a rectangular pulse of height A from t = 0 to t = T, (c) $t\,e^{-2t}$ and $e^{-t}\cos 3t$.

- (a) Linearity + table: $F(s) = 2/s - 3/(s + 2) + 4s/(s^2 + 9)$.
- (b) Pulse = $A\,u(t) - A\,u(t - T)$. Time shift: $F(s) = A/s - A\,e^{-sT}/s = A(1 - e^{-sT})/s$.
- (c) Frequency shift of $t \to 1/s^2$ gives $1/(s + 2)^2$. Frequency shift of $\cos 3t \to s/(s^2 + 9)$ gives $(s + 1)/[(s + 1)^2 + 9]$.

(a) $2/s - 3/(s+2) + 4s/(s^2+9)$ (b) $A(1 - e^{-sT})/s$ (c) $1/(s+2)^2$, $(s+1)/[(s+1)^2+9]$

SOLVED EXAMPLE 3.2 · INITIAL & FINAL VALUE

For $F(s) = \dfrac{5(s + 2)}{s(s + 1)(s + 5)}$, find $f(0^+)$ and $f(\infty)$ without inverting. Then confirm by inverting.

- IVT: $sF(s) = 5(s + 2)/[(s + 1)(s + 5)] \to 5s/s^2 \to 0$ as $s \to \infty \Rightarrow f(0^+) = 0$.
- FVT: poles of $sF(s)$ are -1 and -5 (left half-plane, so FVT is allowed). $s \to 0$: $5 \times 2/(1 \times 5) = 2 \Rightarrow f(\infty) = 2$.
- Check: partial fractions give $f(t) = 2 - 1.25e^{-t} - 0.75e^{-5t}$. At $t = 0$: $2 - 1.25 - 0.75 = 0$ ✓; as $t \to \infty$: 2 ✓.

$f(0^+) = 0$, $f(\infty) = 2$

COMMON MISTAKE

Applying the final value theorem blindly. For $F(s) = 1/(s^2 + 4)$, $sF(s) \to 0$ as $s \to 0$, but $f(t) = \tfrac{1}{2}\sin 2t$ *never settles*. FVT is valid only when $sF(s)$ has no poles on the $j\omega$-axis or in the right half-plane.

3.4 Inverse Laplace by partial fractions

Circuit answers come out as a ratio of polynomials $F(s) = N(s)/D(s)$. To get back to time, split $F(s)$ into simple pieces that are each in the table. Make sure the degree of N is less than that of D (if not, divide first). Then factor D(s) and use the rule for each kind of root:

Type of root of D(s)	Term	Coefficient	Time function
Simple real, s = −p	$K/(s + p)$	K = (s + p)F(s) at s = −p ("cover-up")	$K e^{-pt}$
Repeated, $(s + p)^2$	$K_1/(s + p) + K_2/(s + p)^2$	$K_2 = (s + p)^2F$ at −p; K_1 = d/ds[$(s + p)^2F$] at −p	$K_1e^{-pt} + K_2t\, e^{-pt}$
Complex pair, $(s + a)^2 + \omega^2$	$[B(s + a) + C\omega]/[(s + a)^2 + \omega^2]$	complete the square, compare numerators	$e^{-at}(B \cos \omega t + C \sin \omega t)$

SOLVED EXAMPLE 3.3 · DISTINCT POLES

Find f(t) for $F(s) = \dfrac{s + 5}{(s + 1)(s + 3)}$.

- $F = A/(s + 1) + B/(s + 3)$. Cover-up: $A = (-1 + 5)/(-1 + 3) = 4/2 = 2$; $B = (-3 + 5)/(-3 + 1) = 2/(-2) = -1$.

$f(t) = 2e^{-t} - e^{-3t}, t \geq 0$

SOLVED EXAMPLE 3.4 · REPEATED POLE

Find f(t) for $F(s) = \dfrac{s + 3}{(s + 1)^2(s + 2)}$.

- $F = A/(s + 1) + B/(s + 1)^2 + C/(s + 2)$.
- $B = (s + 3)/(s + 2)$ at $s = -1 = 2/1 = 2$. $C = (s + 3)/(s + 1)^2$ at $s = -2 = 1/1 = 1$.
- A = d/ds$[(s + 3)/(s + 2)]$ at $s = -1 = [(s + 2) - (s + 3)]/(s + 2)^2 = -1/1 = -1$.

$f(t) = -e^{-t} + 2t\, e^{-t} + e^{-2t}$

SOLVED EXAMPLE 3.5 · COMPLEX POLES

Find f(t) for $F(s) = \dfrac{2s + 5}{s^2 + 2s + 5}$.

- Complete the square: $s^2 + 2s + 5 = (s + 1)^2 + 4$, so $a = 1$, $\omega = 2$.
- Rewrite the numerator around (s + 1): $2s + 5 = 2(s + 1) + 3 = 2(s + 1) + 1.5 \times 2$.
- $F = 2 \cdot \dfrac{s + 1}{(s + 1)^2 + 4} + 1.5 \cdot \dfrac{2}{(s + 1)^2 + 4}$.

$f(t) = e^{-t}(2 \cos 2t + 1.5 \sin 2t)$

EXAM TIP

Always check your answer at t = 0 using the initial value theorem — it takes ten seconds. In Example 3.5, lim sF(s) = 2 and f(0) = 2 ✓.

3.5 Solving differential equations with Laplace

The recipe: (1) transform both sides, using the differentiation rule so the initial conditions appear; (2) solve the algebra for Y(s); (3) partial fractions; (4) invert.

SOLVED EXAMPLE 3.6 · KTU STYLE

Solve $y'' + 3y' + 2y = 4u(t)$, with $y(0) = 1$ and $y'(0) = 0$.

- Transform: $[s^2Y - s\cdot 1 - 0] + 3[sY - 1] + 2Y = 4/s$.
- $(s^2 + 3s + 2)Y = 4/s + s + 3 \Rightarrow Y(s) = \dfrac{s^2 + 3s + 4}{s(s + 1)(s + 2)}$.
- Cover-up: at $s = 0$: $4/2 = 2$; at $s = -1$: $(1 - 3 + 4)/(-1 \times 1) = -2$; at $s = -2$: $(4 - 6 + 4)/(-2 \times -1) = 1$.
- Check: $y(0) = 2 - 2 + 1 = 1$ ✓, $y'(0) = 2 - 2 = 0$ ✓.

$$y(t) = 2 - 2e^{-t} + e^{-2t}$$

3.6 Circuits in the s-domain

Instead of writing a differential equation and then transforming it, we can transform the **circuit itself**. Each element gets an s-domain impedance, and each stored initial condition becomes a small extra source. After that, every Module 1–2 tool — KVL, KCL, mesh, node, Thevenin — works unchanged, with s-domain impedances in place of resistances.

ELEMENT LAWS IN THE S-DOMAIN

$$\text{R: } V = R\,I \qquad \text{L: } V(s) = sL\,I(s) - L\,i(0^-) \qquad \text{C: } V(s) = \frac{I(s)}{sC} + \frac{v(0^-)}{s}$$

Impedances: $Z_R = R$, $Z_L = sL$, $Z_C = 1/(sC)$ (put $s = j\omega$ to get back the phasor impedances)

Reading the models. For the inductor, $V = sLI - Li(0^-)$: the stored current acts like a source *helping* the current keep flowing, so its + is at the bottom (where the current leaves). For the capacitor, $V = I/sC + v(0^-)/s$: the stored voltage simply adds a step $v(0^-)/s$ with the same polarity as the capacitor's own charge. Dividing each series source by its impedance (source transformation!) gives the parallel forms.

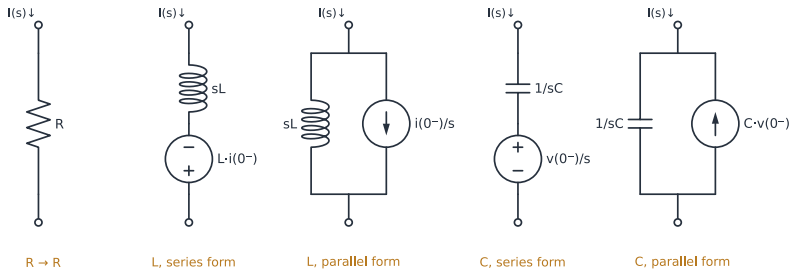


Fig 3.2 s-domain models. Series forms (voltage source) suit mesh/KVL; parallel forms (current source) suit node/KCL.

COMMON MISTAKE

Writing L i(0) as L i(0)/s, or v(0)/s as v(0). Check units: in the s-domain, a voltage source of "step" value V_0 is V_0/s. The inductor's Li(0) is already an impulse-type term (it comes from the derivative rule), so no extra 1/s.

3.7 RC and RL transients

First-order circuits (one energy-storing element) all behave the same way: the response moves **exponentially** from its initial value to its final value, with a speed set by the **time constant τ**.

FIRST-ORDER SHORTCUT — WORKS FOR EVERY RC/RL WITH DC

$$x(t) = x(\infty) + [x(0^+) - x(\infty)]\, e^{-t/\tau} \qquad \tau_{RC} = R_{th}C, \quad \tau_{RL} = L/R_{th}$$

R_{th} = resistance seen by the C or L with independent sources killed.

Physical meaning of τ: in one τ the response covers **63.2 %** of the way to its final value; after 5τ it is within 1 % (99.3 %) and we call it "settled". A big R makes a capacitor charge slowly (less current) but makes an inductor settle fast (L/R small).

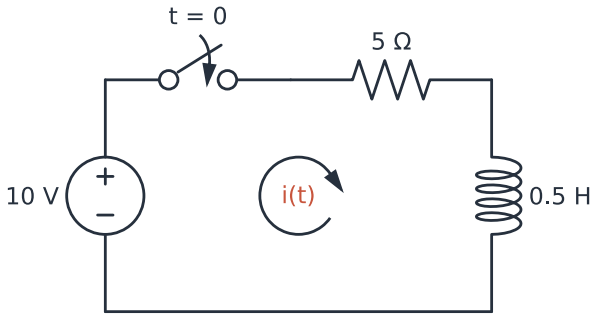


Fig 3.3 RL circuit for Example 3.7.

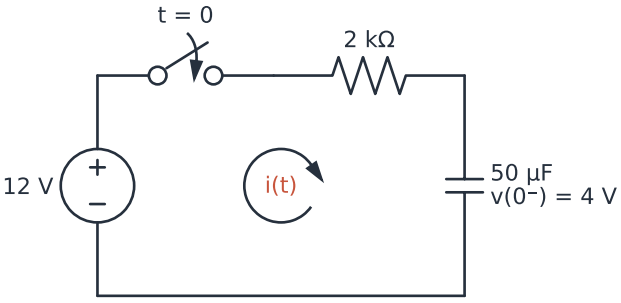


Fig 3.4 RC circuit for Example 3.8.

SOLVED EXAMPLE 3.7 · RL WITH DC (STEP)

In Fig 3.3 the switch closes at t = 0 with no initial current. Find i(t) and $v_L(t)$ using the Laplace transform.

- s-domain KVL: $10/s = (5 + 0.5s)\, I(s) \Rightarrow I(s) = \dfrac{10}{s(0.5s + 5)} = \dfrac{20}{s(s + 10)}$.
- Partial fractions: $2/s - 2/(s + 10) \Rightarrow i(t) = 2(1 - e^{-10t})$ A.
- $v_L = L\, di/dt = 0.5 \times 20e^{-10t} = 10e^{-10t}$ V. $\tau = L/R = 0.1$ s.
- Sanity: at $t = 0^+$, i = 0 (L acts open) and all 10 V appears across L ✓; at $t \to \infty$, L is a short and i = 10/5 = 2 A ✓.

$i(t) = 2(1 - e^{-10t})$ A, $v_L(t) = 10e^{-10t}$ V, $\tau = 0.1$ s (curves in Fig 3.5)

SOLVED EXAMPLE 3.8 · RC WITH AN INITIAL CONDITION

In Fig 3.4 the capacitor already holds $v(0^-) = 4$ V when the switch closes. Find i(t) and $v_C(t)$.

- Use the series model: 1/sC in series with 4/s. KVL: $12/s = 2000\, I + I/(50 \times 10^{-6}\ s) + 4/s$.
- $I(s)(2000 + 20000/s) = 8/s \Rightarrow I(s) = \dfrac{8}{2000s + 20000} = \dfrac{0.004}{s + 10}$.
- $i(t) = 4e^{-10t}$ mA. $\tau = RC = 2000 \times 50\ \mu F = 0.1$ s.
- $v_C = 12 - 2000\, i = 12 - 8e^{-10t}$ V (starts at 4 V ✓, ends at 12 V ✓). Same as the shortcut: $12 + (4 - 12)e^{-t/\tau}$.

$i(t) = 4e^{-10t}$ mA, $v_C(t) = 12 - 8e^{-10t}$ V

SOLVED EXAMPLE 3.9 · IMPULSE RESPONSE OF RC

A series RC (R = 1 kΩ, C = 1 mF, uncharged) is driven by $v_s = \delta(t)$ V·s. Find $v_C(t)$ and i(t).

- Voltage divider in s: $V_C(s) = 1 \cdot \dfrac{1/sC}{R + 1/sC} = \dfrac{1/RC}{s + 1/RC}$, with $\tau = RC = 1$ s.
- $v_C(t) = (1/\tau)\, e^{-t/\tau} = e^{-t}$ V — the capacitor jumps to 1 V at $t = 0^+$, then discharges.
- $I(s) = 1/(R + 1/sC) = (1/R)[1 - (1/\tau)/(s + 1/\tau)] \Rightarrow i(t) = 1$ mA·s $\times\ \delta(t) - 1\, e^{-t}$ mA.
- Why the jump? The impulse pushes a charge $q = 1/R = 1$ mC in zero time, so $v = q/C = 1$ V. (An impulse is the one input that *can* change a capacitor voltage instantly.)

$v_C(t) = e^{-t}$ V, $i(t) = 10^{-3}\delta(t) - 10^{-3}e^{-t}$ A

KEY IDEA — IMPULSE VS STEP

The impulse response is the **derivative** of the step response (because δ = du/dt). If you know one, you know the other. In s-terms: impulse response = H(s), step response = H(s)/s.

Sinusoidal input

With an AC source switched in at t = 0, the response has two parts: a **steady-state** sinusoid (what phasors from Module 1 would predict) plus a decaying **transient** that makes the total satisfy the initial condition.

SOLVED EXAMPLE 3.10 · RL WITH SINUSOIDAL SOURCE

v(t) = 10 sin 5t V is switched at t = 0 onto a series R = 5 Ω, L = 1 H (no initial current). Find i(t).

▸ $I(s) = \frac{V(s)}{R + sL} = \frac{10 \times 5}{(s^2 + 25)(s + 5)}$.

▸ $A/(s + 5)$: $A = 50/(25 + 25) = 1$. The rest is $(Bs + C)/(s^2 + 25)$: matching s^2 terms gives $B = -1$; matching constants, $50 = 25 + 5C \Rightarrow C = 5$.

▸ $i(t) = e^{-5t} - \cos 5t + \sin 5t = e^{-5t} + 1.414 \sin(5t - 45°)$ A.

▸ Phasor check: $|Z| = |5 + j5| = 7.07\ \Omega$, $\theta = 45°$, so steady amplitude $10/7.07 = 1.414$ A lagging 45° √. At t = 0: $1 + 1.414 \sin(-45°) = 0$ √.

$i(t) = 1.414 \sin(5t - 45°) + e^{-5t}$ A

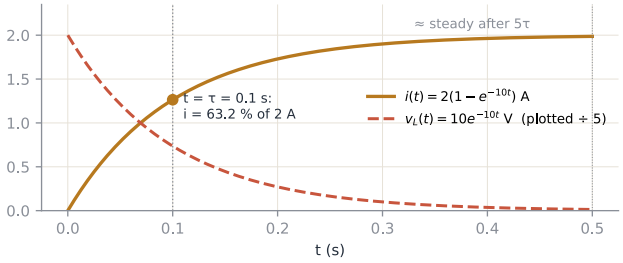


Fig 3.5 Example 3.7: current rise and inductor voltage decay.

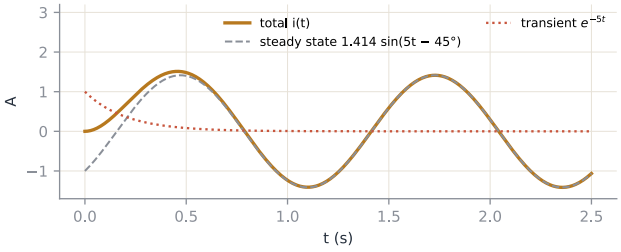


Fig 3.6 Example 3.10: transient + steady state = total current.

EXAM TIP

If the switch closes at the instant when the source phase φ equals the impedance angle θ (i.e. $\sin(\varphi - \theta) = 0$), there is **no transient at all** — the current starts directly in steady state. KTU sometimes asks for this switching angle.

INTERVIEW CORNER

Q. What is the time constant, physically?

A. The time the response would take to reach its final value if it kept its initial slope. In practice: 63.2 % of the change after one τ, ~99 % after 5τ.

Q. A capacitor is charged to 5 V and suddenly shorted with a wire. What happens?

A. Ideally an infinite current impulse ($\tau = 0$). Really, the wire's and capacitor's tiny resistance and inductance limit it — that's why large capacitors are discharged through a resistor.

Q. Why does opening a switch in an inductive circuit cause a spark?

A. The inductor tries to keep its current flowing; $v = L\,di/dt$ becomes huge as $di/dt \to -\infty$, so the air gap breaks down. Flyback diodes across relay coils prevent this.

3.8 RLC transients and damping

With both L and C, energy can slosh back and forth between the magnetic field and the electric field — like a mass on a spring. The resistor decides how quickly this oscillation dies out. For a series RLC with a DC step V, the s-domain current is

SERIES RLC (ZERO INITIAL CONDITIONS)

$$I(s) = \frac{V/s}{R + sL + 1/sC} = \frac{V/L}{s^2 + (R/L)s + 1/LC}$$

$$\text{Roots: } s_{1,2} = -\alpha \pm \sqrt{(\alpha^2 - \omega_0^2)}, \quad \alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{(LC)}}, \quad \zeta = \frac{\alpha}{\omega_0} = \frac{R}{2}\sqrt{(C/L)}$$

Case	Condition	Roots	Shape of response
Overdamped	$\alpha > \omega_0$ ($\zeta > 1$), $R > 2\sqrt{(L/C)}$	two real, distinct	sum of two decaying exponentials, sluggish, no overshoot
Critically damped	$\alpha = \omega_0$ ($\zeta = 1$), $R = 2\sqrt{(L/C)}$	real, repeated ($-\alpha$)	$(A + Bt)e^{-\alpha t}$; fastest without overshoot
Underdamped	$\alpha < \omega_0$ ($\zeta < 1$)	$-\alpha \pm j\omega_d$, $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$	decaying sinusoid $e^{-\alpha t} \sin \omega_d t$ (ringing)
Undamped	$R = 0$	$\pm j\omega_0$	sustained oscillation at ω_0

For a **parallel** RLC, everything is the same except $\alpha = 1/(2RC)$ — so a *larger* R means *less* damping there.

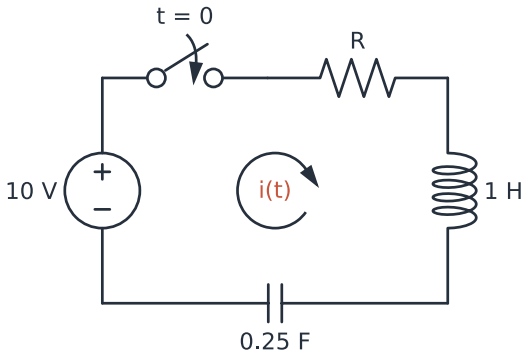


Fig 3.7 Series RLC for Examples 3.11 and 3.12.

SOLVED EXAMPLE 3.11 · ALL THREE DAMPING CASES

In Fig 3.7 (L = 1 H, C = 0.25 F, initially relaxed), the 10 V source is switched in at t = 0. Find i(t) for R = 5 Ω, 4 Ω and 2 Ω.

▸ $I(s) = (10/s)/(R + s + 4/s) = 10/(s^2 + Rs + 4)$. $\omega_0 = 1/\sqrt{(0.25)} = 2$ rad/s; critical $R = 2\sqrt{(L/C)} = 2\sqrt{4} = 4\ \Omega$.

▸ **R = 5 Ω** (overdamped): $s^2 + 5s + 4 = (s + 1)(s + 4)$. $I = (10/3)[1/(s + 1) - 1/(s + 4)] \Rightarrow i = (10/3)(e^{-t} - e^{-4t})$ A.

▸ **R = 4 Ω** (critical): $s^2 + 4s + 4 = (s + 2)^2 \Rightarrow i = 10t\, e^{-2t}$ A.

▸ **R = 2 Ω** (underdamped): $s^2 + 2s + 4 = (s + 1)^2 + 3$, $\omega_d = \sqrt{3} = 1.732$ rad/s. $I = (10/\sqrt{3})\cdot\sqrt{3}/[(s + 1)^2 + 3] \Rightarrow i = 5.774\, e^{-t} \sin 1.732t$ A.

▸ All start at $i(0^+) = 0$ (L is open at first) and end at 0 (C blocks DC) √.

5 Ω: $3.333(e^{-t} - e^{-4t})$; 4 Ω: $10te^{-2t}$; 2 Ω: $5.774e^{-t} \sin 1.732t$ (A)

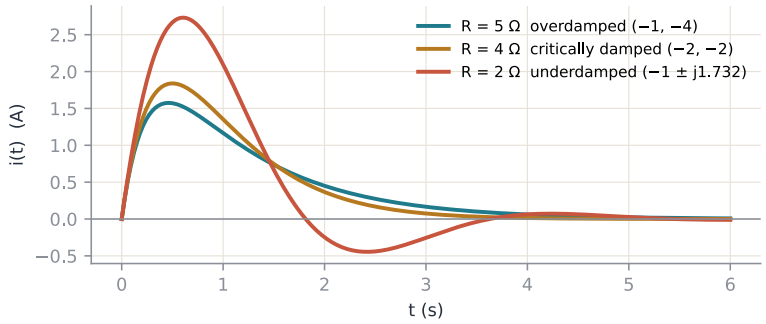


Fig 3.8 The three responses of Example 3.11. Only the underdamped one dips below zero (rings).

SOLVED EXAMPLE 3.12 · RLC IMPULSE RESPONSE

Same circuit with R = 2 Ω, but the source is a unit impulse δ(t). Find i(t).

▸ $V(s) = 1 \Rightarrow I(s) = 1/(2 + s + 4/s) = \dfrac{s}{s^2 + 2s + 4} = \dfrac{(s + 1) - 1}{(s + 1)^2 + 3}$.

▸ $i(t) = e^{-t}\cos 1.732t - (1/1.732)\,e^{-t}\sin 1.732t$. (Check: it is the derivative of the step answer divided by 10 √.)

$i(t) = e^{-t}(\cos 1.732t - 0.577 \sin 1.732t)$ A

KEY IDEA — ANY INPUT, SAME DENOMINATOR

Step, impulse or sinusoid — the **denominator** $s^2 + (R/L)s + 1/LC$ (the characteristic equation) is the same, so the *natural* part of the response always has the same shape. The input only changes the numerator and adds a forced (steady-state) part. For a sinusoidal input, find the forced part by phasors and the natural part from these roots.

INTERVIEW CORNER

Q. Why do PCB designers worry about "ringing" on fast digital edges?

A. Trace inductance and pin capacitance form an underdamped RLC. A series termination resistor raises damping towards $\zeta \approx 1$ and kills the overshoot.

Q. Define quality factor for a series RLC.

A. $Q = \omega_0 L/R = 1/(2\zeta)$. High Q ⇒ low damping, sharp resonance, long ringing.

3.9 Low-pass and high-pass RC circuits

Put a resistor and capacitor in series. Take the output **across C** and you have a **low-pass** circuit: the capacitor cannot follow fast changes, so it smooths them out. Take the output **across R** and you have a **high-pass** circuit: it passes sudden changes and blocks steady (DC) levels. These two little circuits appear everywhere — RC delays on reset pins, coupling capacitors between amplifier stages, anti-aliasing filters.

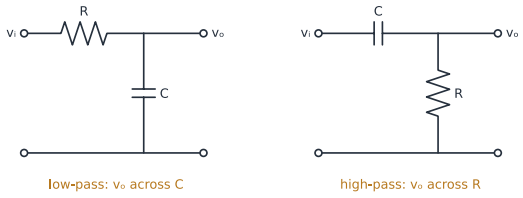


Fig 3.9 RC low-pass (integrator-like) and high-pass (differentiator-like) circuits.

STEP RESPONSE (INPUT V AT T = 0, T = RC)

$$\text{Low-pass: } v_o = V(1 - e^{-t/\tau}) \qquad \text{High-pass: } v_o = V\,e^{-t/\tau}$$

$$\text{Transfer: } H_{LP}(s) = \frac{1}{1 + sRC}, \quad H_{HP}(s) = \frac{sRC}{1 + sRC}, \quad f_c = \frac{1}{2\pi RC}$$

$$\text{Rise time (10 \%→90 \%): } t_r = 2.2\,RC = 0.35/f_c$$

Response to a pulse

A pulse of width t_p is a step up at 0 and a step down at t_p, so its response is the step response minus a delayed step response. The shape depends entirely on how τ compares with t_p:

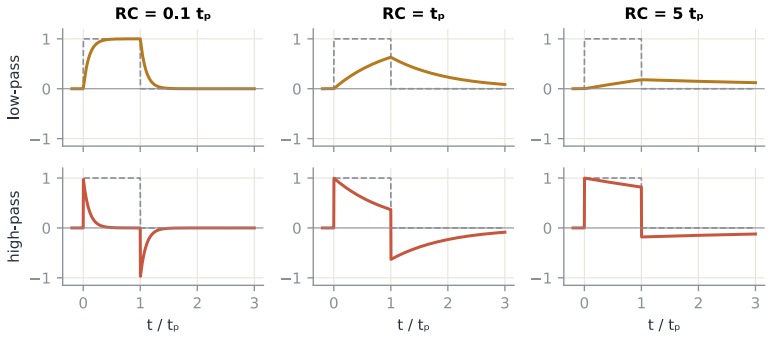


Fig 3.10 Pulse response (dashed = input). Low-pass: small τ keeps the shape, large τ integrates. High-pass: small τ gives spikes (differentiator), large τ passes the pulse with slight tilt.

τ vs t_p	Low-pass output	High-pass output
$\tau \ll t_p$	faithful copy, slightly rounded edges	positive and negative spikes at the edges — acts as a **differentiator**
$\tau \approx t_p$	rises to 63 % then decays	decays to 37 %, then jumps negative by V
$\tau \gg t_p$	small ramp — acts as an **integrator**	almost flat top with a small **tilt** $\approx t_p/\tau$, small negative undershoot

SOLVED EXAMPLE 3.13 · PULSE THROUGH LPF AND HPF

A 5 V, 2 ms pulse is applied to an RC circuit with R = 1 kΩ and C = 1 µF. Find the output at t = 2 ms and t = 4 ms when the output is taken (a) across C, (b) across R. Also find the rise time and cut-off frequency.

▸ $\tau = RC = 1$ ms, so $t_p = 2\tau$.

▸ (a) Low-pass: $v_o(2\text{ ms}) = 5(1 - e^{-2}) = 4.323$ V. After the pulse ends it decays: $v_o(4\text{ ms}) = 4.323\, e^{-2} = 0.585$ V.

▸ (b) High-pass: $v_R = v_{in} - v_C$. Just before 2 ms: $5 - 4.323 = 0.677$ V. At 2 ms the input drops by 5 V and C can't change, so v_R jumps to $0.677 - 5 = -4.323$ V, then decays: at 4 ms, $-4.323e^{-2} = -0.585$ V.

▸ $t_r = 2.2 \times 1$ ms $= 2.2$ ms; $f_c = 1/(2\pi \times 10^{-3}) = 159.2$ Hz.

LPF: 4.323 V, 0.585 V HPF: 0.677 V → −4.323 V, −0.585 V t_r = 2.2 ms, f_c = 159.2 Hz

COMMON MISTAKE

Forgetting that the high-pass output goes **negative** after the pulse ends. The capacitor still holds its charge, so when the input falls, the whole drop appears across R.

INTERVIEW CORNER

Q. Relate rise time and bandwidth.

A. For a first-order system $t_r = 2.2\tau$ and $f_{3dB} = 1/2\pi\tau$, so $t_r \times BW \approx 0.35$. A 100 MHz scope probe can show edges no faster than about 3.5 ns.

Q. When does an RC high-pass behave as a differentiator?

A. When $\tau \ll$ the signal period/pulse width: $v_R \approx RC \cdot dv_{in}/dt$. Likewise RC low-pass ≈ integrator when $\tau \gg$ the period.

Q. Why is a coupling capacitor between amplifier stages a high-pass filter, and what does that do to square waves?

A. It blocks DC bias and passes AC; with a finite τ the flat tops of a square wave tilt (sag). Make $\tau \gg$ the lowest period to keep tilt small.

✓ Module 3 — Quick revision

Must-know pairs: $\delta \to 1$, $u \to 1/s$, $t \to 1/s^2$, $e^{-at} \to 1/(s+a)$, $\sin \omega t \to \omega/(s^2+\omega^2)$, $\cos \omega t \to s/(s^2+\omega^2)$.

Shifts: $f(t-T)u(t-T) \to e^{-sT}F(s)$; $e^{-at}f(t) \to F(s+a)$. **Derivative:** $sF - f(0^-)$; **integral:** F/s.

IVT / FVT: $f(0^+) = \lim_{s\to\infty} sF$; $f(\infty) = \lim_{s\to 0} sF$ (stable only).

Partial fractions: cover-up for simple poles; derivative for repeated; complete the square for complex.

s-models: $L \to sL$ with source $Li(0^-)$; $C \to 1/sC$ with source $v(0^-)/s$.

At t = 0⁺: L keeps its current, C keeps its voltage. At DC steady state: L short, C open.

First order: $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$; $\tau = RC$ or L/R.

Series RLC: $\alpha = R/2L$, $\omega_0 = 1/\sqrt{LC}$; critical $R = 2\sqrt{(L/C)}$. Parallel: $\alpha = 1/2RC$.

RC filters: $f_c = 1/2\pi RC$, $t_r = 2.2RC$; HPF ≈ differentiator if $\tau \ll t_p$, LPF ≈ integrator if $\tau \gg t_p$.

Practice — short questions (3 marks)

1. Define the Laplace transform and find the transform of $e^{-at}\cos \omega t$ from first principles.
2. State and prove the time-shifting theorem. Find $\mathscr{L}\{u(t-3)\}$. [e^{-3s}/s]
3. State the initial and final value theorems. Find $f(0^+)$, $f(\infty)$ for $F(s) = (s+4)/[s(s+2)]$. [1, 2]
4. Draw the s-domain equivalents of an inductor and a capacitor with initial conditions.
5. Find the time constant of an RL circuit with R = 20 Ω, L = 0.2 H, and the current after one τ for a 10 V step. [10 ms, 0.316 A]
6. Define overdamped, critically damped and underdamped responses of an RLC circuit.
7. Find the rise time and cut-off frequency of an RC low-pass circuit with R = 10 kΩ, C = 0.1 μF. [2.2 ms, 159.2 Hz]

Practice — long questions (9/14 marks)

1. Find the inverse Laplace transform of $(3s+7)/[(s+1)(s+2)(s+3)]$. [Ans: $2e^{-t} - e^{-2t} - e^{-3t}$]
2. Solve $y'' + 4y' + 3y = 6u(t)$, $y(0) = 0$, $y'(0) = 2$ using the Laplace transform. [Ans: $y = 2 - 2e^{-t}$]
3. In Fig 3.3, after the current has reached 2 A, the source is replaced by a short (switch to a second position). Find i(t). [Ans: $2e^{-10t}$ A]
4. Derive the step response of a series RLC circuit; discuss the three damping cases.
5. In Fig 3.7 with R = 2 Ω, find the voltage across the capacitor for the 10 V step. [Ans: $10 - 10e^{-t}(\cos 1.732t + 0.577 \sin 1.732t)$ V]
6. Derive and sketch the RC low-pass and high-pass pulse responses for $\tau \ll t_p$ and $\tau \gg t_p$.

04

Network Functions & Two-Port Networks

> *This is the language of amplifiers, filters and RF blocks: a transistor's h-parameters, an RF chip's S/ABCD data sheet, a filter's poles and zeros. Interviewers love "what does a pole in the right half-plane mean?"*

SYLLABUS CHECKLIST

- ☐ Network functions: driving-point and transfer functions
- ☐ Poles and zeros, their significance, pole-zero plots
- ☐ Time-domain behaviour from pole locations
- ☐ Restrictions on pole and zero locations
- ☐ Two-port networks: Z, Y, h (hybrid) and ABCD (transmission) parameters
- ☐ Conditions for reciprocity and symmetry
- ☐ Interrelationships between the parameter sets
- ☐ Series, parallel and cascade interconnection of two-ports

11 hours · CO4 · Analyse networks using network functions and two-port parameters

4.1 Network functions

In Module 3 we found that, with zero initial conditions, every s-domain response is simply *(something about the circuit) × (input)*. That "something about the circuit" is the **network function**:

DEFINITION

$$H(s) = \frac{\text{Response transform } R(s)}{\text{Excitation transform } E(s)} \quad \text{(all initial conditions zero)} \quad = \frac{N(s)}{D(s)}$$

It is a property of the network alone — not of the input. Once you know H(s), the response to any input is R(s) = H(s)E(s), and the impulse response is simply $h(t) = \mathscr{L}^{-1}\{H(s)\}$.

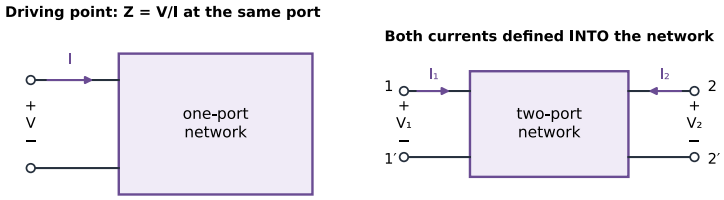


Fig 4.1 A port is a pair of terminals where the current entering one equals the current leaving the other. Standard convention: currents flow *into* the + terminal.

Type	Excitation and response	Name and symbol
Driving-point (same port)	I_1 in, V_1 out	driving-point impedance $Z_{11}(s) = V_1/I_1$
	V_1 in, I_1 out	driving-point admittance $Y_{11}(s) = I_1/V_1 = 1/Z_{11}$
Transfer (different ports)	V_1 in, V_2 out	voltage transfer ratio $G_{21}(s) = V_2/V_1$
	I_1 in, I_2 out	current transfer ratio $\alpha_{21}(s) = I_2/I_1$
	I_1 in, V_2 out	transfer impedance $Z_{21}(s) = V_2/I_1$
	V_1 in, I_2 out	transfer admittance $Y_{21}(s) = I_2/V_1$

THINK OF IT LIKE

Knocking on a wall. The *driving-point* function is how the wall pushes back on your own hand. A *transfer* function is how loud the knock sounds in the next room. Same wall, different question.

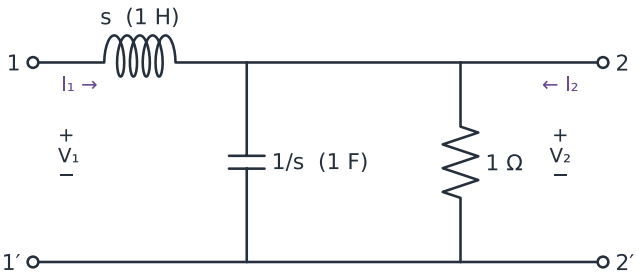


Fig 4.2 s-domain ladder for Examples 4.1 and 4.2 (port 2 open).

SOLVED EXAMPLE 4.1 · KTU STYLE

For the network of Fig 4.2 (port 2 open), find the driving-point impedance $Z_{11}(s)$, the voltage transfer ratio V_2/V_1 and the transfer impedance V_2/I_1.

- Shunt branch: $(1/s) \parallel 1 = \dfrac{(1/s)(1)}{1/s + 1} = \dfrac{1}{s + 1}$.
- $Z_{11}(s) = s + \dfrac{1}{s + 1} = \dfrac{s^2 + s + 1}{s + 1}$.
- Voltage divider: $V_2/V_1 = \dfrac{1/(s + 1)}{(s^2 + s + 1)/(s + 1)} = \dfrac{1}{s^2 + s + 1}$.
- $V_2 = I_1 \times$ (shunt impedance) $\Rightarrow V_2/I_1 = 1/(s + 1)$.

$Z_{11} = (s^2 + s + 1)/(s + 1)$; $G_{21} = 1/(s^2 + s + 1)$; $Z_{21} = 1/(s + 1)$

That $G_{21} = 1/(s^2 + s + 1)$ is a famous one: a second-order **Butterworth low-pass filter** with cut-off 1 rad/s. Three components, and you have built a real filter.

4.2 Poles, zeros and pole-zero plots

Factor numerator and denominator of a network function:

FACTORED FORM

$$H(s) = K\,\frac{(s - z_1)(s - z_2)\ldots(s - z_m)}{(s - p_1)(s - p_2)\ldots(s - p_n)}$$

- **Zeros** z_k: values of s where H(s) = 0 — drawn as ○.
- **Poles** p_k: values of s where H(s) → ∞ — drawn as ×.
- K is the scale factor. Poles, zeros and K together describe H(s) **completely**.

KEY IDEA — SIGNIFICANCE

Poles are the natural frequencies of the network: the denominator D(s) is the characteristic equation of Module 3, so each pole p gives a term e^{pt} in the natural response. **Zeros** are frequencies the network blocks (zero transmission) — e.g. a notch filter has zeros on the jω-axis at the frequency it rejects. Zeros affect the *amplitudes* of the response terms, poles decide their *shapes*.

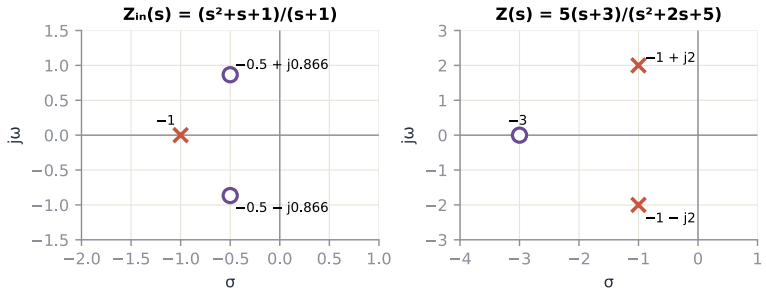


Fig 4.3 Pole-zero plots for Example 4.1 (left) and Example 4.2 (right).

SOLVED EXAMPLE 4.2 · FROM A POLE-ZERO PLOT

(a) Draw the pole-zero plot of $Z_{11}(s)$ from Example 4.1. (b) A driving-point impedance has poles at −1 ± j2, a zero at −3 and Z(0) = 3 Ω. Find Z(s) and its impulse response.

- (a) Zeros: $s^2 + s + 1 = 0 \Rightarrow s = -0.5 \pm j0.866$. Pole: $s = -1$. (Fig 4.3 left.)
- (b) $Z(s) = K(s + 3)/[(s + 1 - j2)(s + 1 + j2)] = K(s + 3)/(s^2 + 2s + 5)$.
- $Z(0) = 3K/5 = 3 \Rightarrow K = 5 \Rightarrow Z(s) = 5(s + 3)/(s^2 + 2s + 5)$.
- Write $s + 3 = (s + 1) + 2$: $Z(s) = 5\dfrac{s + 1}{(s + 1)^2 + 4} + 5\dfrac{2}{(s + 1)^2 + 4}$.

$Z(s) = 5(s + 3)/(s^2 + 2s + 5)$; $z(t) = 5e^{-t}(\cos 2t + \sin 2t)$

Time-domain behaviour from pole locations

Because each pole $p = \sigma + j\omega$ contributes $e^{\sigma t}\cdot(\cos/\sin \omega t)$, you can *read the response off the s-plane* without any algebra:

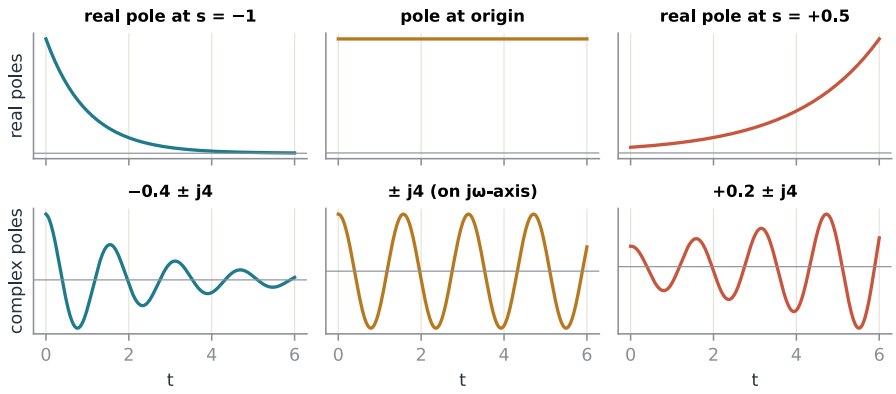


Fig 4.4 Impulse-response term produced by each pole location. Left column: stable; middle: marginal; right: unstable.

Pole location	Response term	Stability
Negative real axis, s = −a	decaying exponential e^{-at} (faster the further left)	stable
Complex pair in LHP, $-\sigma \pm j\omega$	damped oscillation $e^{-\sigma t}\cos \omega t$	stable
Simple pole at origin / simple pair on jω-axis	constant / sustained sinusoid	marginally stable
Repeated poles on jω-axis	$t \cdot \cos \omega t$ — grows without bound	unstable
Any pole in the RHP ($\sigma > 0$)	growing exponential / growing oscillation	unstable

Distance from the jω-axis sets the **decay rate**; height above the real axis sets the **oscillation frequency**. Poles close to the jω-axis "ring" for a long time — that is a high-Q circuit.

INTERVIEW CORNER

Q. What does a pole in the right half-plane mean physically?

A. A natural response that grows exponentially — the circuit is unstable. Purely passive RLC networks cannot do this; you need active elements (an oscillator deliberately places poles on, or just right of, the jω-axis to start oscillating).

Q. Can a zero lie in the right half-plane? What does it do?

A. Yes, in a transfer function (not in a passive driving-point function). An RHP zero gives "non-minimum phase" behaviour — the step response first moves the wrong way. The Miller capacitance in a CE/CS amplifier creates one.

4.3 Restrictions on pole and zero locations

Not every rational function can be built from real R, L, C. Passive networks obey strict rules — useful for spotting impossible answers.

Driving-point functions (Z_{11}, Y_{11})	Transfer functions (G_{21}, α_{21}, Z_{21}, Y_{21})

1. Coefficients of N(s) and D(s) real and **positive**.	1. Coefficients real; those of D(s) positive, N(s) may have negative ones.
2. Complex poles and zeros occur in conjugate pairs.	2. Complex poles and zeros occur in conjugate pairs.
3. Real parts of **poles and zeros** ≤ 0; those on the jω-axis must be simple.	3. Real parts of **poles** ≤ 0, jω-axis poles simple. **Zeros may lie anywhere**, even in the RHP.
4. No missing terms in N or D between highest and lowest powers, unless all even or all odd.	4. D(s) has no missing terms (unless all even/odd); N(s) may have missing terms and may even be a constant.
5. Degrees of N and D differ by at most 1.	5. Degree of N can be as small as zero, independent of D.
6. Lowest powers of N and D differ by at most 1.	6. For G_{21}, α_{21}: deg N ≤ deg D. For Z_{21}, Y_{21}: deg N ≤ deg D + 1.

Why rule 5 (driving point)? At very high or very low frequency every passive impedance must look like an R, an L ($\propto s$) or a C ($\propto 1/s$) — so Z can grow or fall at most like $s^{\pm 1}$.

SOLVED EXAMPLE 4.3 · VALID OR NOT?

Which of these can be the driving-point impedance of a passive network? (a) $(s^2 + 2s + 2)/(s + 3)$ (b) $(s + 2)/(s^3 + 4s^2 + 5s + 2)$ (c) $(s - 1)/(s + 2)$ (d) $(s^2 + 4)/[s(s^2 + 1)]$ (e) $1/(s^2 + 4)^2$

- (a) Positive coefficients, degrees differ by 1, zeros $-1 \pm j1$ and pole -3 in LHP ⇒ **valid**.
- (b) Degrees 1 and 3 differ by 2 ⇒ **not valid** (it could be a transfer function).
- (c) Zero at $s = +1$ in the RHP and a negative coefficient ⇒ **not valid**.
- (d) Simple jω-axis poles $(0, \pm j1)$ and zeros $(\pm j2)$, alternating, degrees differ by 1 ⇒ **valid** (a pure LC network).
- (e) Repeated poles at $\pm j2$ and degree difference 4 ⇒ **not valid**.

Valid: (a) and (d) only

EXAM TIP

For "check whether this is a valid driving-point function" questions, go through the six rules in order and quote the first one that fails. One clear reason earns full marks.

4.4 Two-port networks: Z and Y parameters

Many circuits have an **input port** and an **output port** — an amplifier, a filter, a transformer, a transmission line. We often don't care what's inside; we want a compact "data sheet" that tells us how V_1, I_1, V_2, I_2 relate. Four variables, two equations: choose any two as independent, and you get a parameter set. There are six choices; the syllabus uses four.

THINK OF IT LIKE

Describing a car to a buyer: you can quote "0–100 km/h time and top speed" or "power and weight" — different numbers, same car. Z, Y, h and ABCD are different but equivalent descriptions of the same two-port.

Z (open-circuit impedance) parameters

Z PARAMETERS

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \qquad V_2 = Z_{21}I_1 + Z_{22}I_2$$

$$Z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0} \quad Z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0} \quad Z_{12} = \frac{V_1}{I_2}\Big|_{I_1=0} \quad Z_{22} = \frac{V_2}{I_2}\Big|_{I_1=0}$$

"Open-circuit" because each is measured with the *other* port open (its current zero). Z_{11}, Z_{22} are driving-point impedances; Z_{12}, Z_{21} are transfer impedances.

Y (short-circuit admittance) parameters

Y PARAMETERS

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \qquad I_2 = Y_{21}V_1 + Y_{22}V_2$$

$$Y_{11} = \frac{I_1}{V_1}\Big|_{V_2=0} \quad Y_{21} = \frac{I_2}{V_1}\Big|_{V_2=0} \quad Y_{12} = \frac{I_1}{V_2}\Big|_{V_1=0} \quad Y_{22} = \frac{I_2}{V_2}\Big|_{V_1=0}$$

$$[Y] = [Z]^{-1}$$

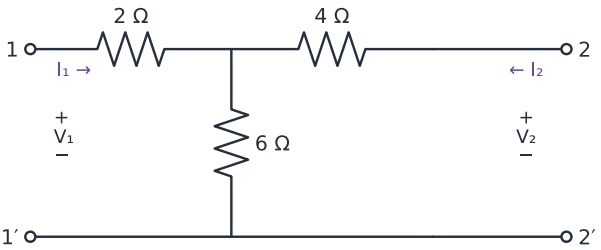


Fig 4.5 T network (Examples 4.4, 4.6, 4.7).

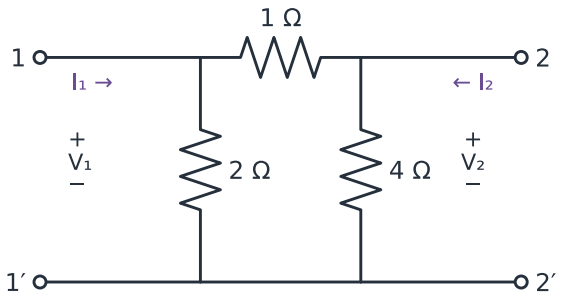


Fig 4.6 π network (Example 4.5).

SOLVED EXAMPLE 4.4 · Z PARAMETERS OF A T

Find the Z parameters of the T network in Fig 4.5.

▸ Port 2 open ($I_2 = 0$): I_1 flows through 2 Ω and 6 Ω. $Z_{11} = 2 + 6 = 8$ Ω. No current in 4 Ω, so $V_2 = 6I_1 \Rightarrow Z_{21} = 6$ Ω.

▸ Port 1 open ($I_1 = 0$): $Z_{22} = 4 + 6 = 10$ Ω; $V_1 = 6I_2 \Rightarrow Z_{12} = 6$ Ω.

▸ General T (series arms Z_a, Z_b, shunt Z_c): $Z_{11} = Z_a + Z_c$, $Z_{22} = Z_b + Z_c$, $Z_{12} = Z_{21} = Z_c$.

[Z] = [8 6; 6 10] Ω (ΔZ = 80 − 36 = 44)

SOLVED EXAMPLE 4.5 · Y PARAMETERS OF A Π

Find the Y parameters of the π network in Fig 4.6.

▸ Admittances: $Y_A = 1/2 = 0.5$ S (shunt at port 1), $Y_C = 1$ S (series), $Y_B = 1/4 = 0.25$ S (shunt at port 2).

▸ Short port 2 ($V_2 = 0$): $Y_{11} = Y_A + Y_C = 1.5$ S. Current through the series 1 Ω is $V_1/1$ flowing *out of* port 2 terminal, so $I_2 = -V_1 \Rightarrow Y_{21} = -1$ S.

▸ Short port 1: $Y_{22} = Y_B + Y_C = 1.25$ S, $Y_{12} = -1$ S.

[Y] = [1.5 −1; −1 1.25] S

COMMON MISTAKE

Getting the sign of Y_{12}/Y_{21} wrong. In a passive π, the transfer admittance is **negative** ($-Y_{series}$) because I_2 is defined *into* port 2 while the actual current flows out.

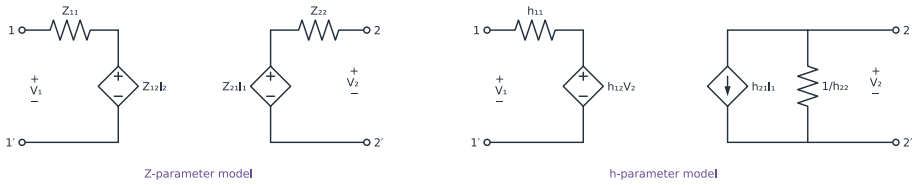


Fig 4.7 Equivalent circuits built straight from the equations: Z model (two CCVSs) and h model (VCVS + CCCS — the familiar BJT small-signal model).

4.5 h (hybrid) and ABCD (transmission) parameters

h parameters

H PARAMETERS

$$V_1 = h_{11}I_1 + h_{12}V_2 \qquad I_2 = h_{21}I_1 + h_{22}V_2$$

$$h_{11} = \frac{V_1}{I_1}\Big|_{V_2=0}\ (\Omega) \quad h_{21} = \frac{I_2}{I_1}\Big|_{V_2=0} \quad h_{12} = \frac{V_1}{V_2}\Big|_{I_1=0} \quad h_{22} = \frac{I_2}{V_2}\Big|_{I_1=0}\ (\text{S})$$

"Hybrid" because the units are mixed: h_{11} is the input impedance (output shorted), h_{21} the forward current gain, h_{12} the reverse voltage gain, h_{22} the output admittance (input open). These suit transistors perfectly: a BJT's base input is low-impedance and its collector output high-impedance, so shorting the output and opening the input are easy measurements. In BJT notation $h_{11} = h_{ie}$, $h_{21} = h_{fe}$ (= β), and so on.

ABCD (transmission, T) parameters

ABCD PARAMETERS

$$V_1 = A\,V_2 - B\,I_2 \qquad I_1 = C\,V_2 - D\,I_2$$

$$A = \frac{V_1}{V_2}\Big|_{I_2=0} \quad C = \frac{I_1}{V_2}\Big|_{I_2=0}\ (\text{S}) \quad B = -\frac{V_1}{I_2}\Big|_{V_2=0}\ (\Omega) \quad D = -\frac{I_1}{I_2}\Big|_{V_2=0}$$

ABCD relates the **input** to the **output** — ideal for transmission lines and for chains of networks. The minus sign appears because the output current that actually flows out to a load is $-I_2$.

SOLVED EXAMPLE 4.6 · H PARAMETERS DIRECTLY

Find the h parameters of the T network of Fig 4.5 by the defining tests.

- Short port 2 ($V_2 = 0$): input sees $2 + (6 \parallel 4) = 2 + 2.4 = 4.4\ \Omega \Rightarrow h_{11} = 4.4\ \Omega$.
- Current divider: current into the short $= I_1 \times 6/(6+4) = 0.6I_1$ flowing *out* of port 2 $\Rightarrow I_2 = -0.6I_1$ $\Rightarrow h_{21} = -0.6$.
- Open port 1 ($I_1 = 0$): $I_2 = V_2/10 \Rightarrow h_{22} = 0.1$ S; $V_1 = 6I_2 = 0.6V_2 \Rightarrow h_{12} = 0.6$.

$h_{11} = 4.4\ \Omega$, $h_{12} = 0.6$, $h_{21} = -0.6$, $h_{22} = 0.1$ S

SOLVED EXAMPLE 4.7 · ABCD PARAMETERS

Find the ABCD parameters of Fig 4.5 and check AD − BC.

- Port 2 open: $V_1 = 8I_1$, $V_2 = 6I_1 \Rightarrow A = 8/6 = 1.333$, $C = I_1/V_2 = 1/6 = 0.1667$ S.
- Port 2 shorted: from Example 4.6, $V_1 = 4.4I_1$, $I_2 = -0.6I_1 \Rightarrow B = -V_1/I_2 = 4.4/0.6 = 7.333\ \Omega$, $D = -I_1/I_2 = 1/0.6 = 1.667$.
- $AD - BC = (4/3)(5/3) - (22/3)(1/6) = 20/9 - 11/9 = 1$ ✓ (reciprocal, as expected for R-only).

$A = 4/3$, $B = 22/3\ \Omega$, $C = 1/6$ S, $D = 5/3$; $AD - BC = 1$

INTERVIEW CORNER

Q. Why are h parameters used for BJTs rather than Z or Y?

A. h_{11} needs the output shorted and h_{22} needs the input open — both easy for a BJT (low input impedance, high output impedance). Measuring Z_{22} with a truly open input, or Y_{11} with a shorted high-impedance output, is awkward. Also h_{fe} directly equals current gain β.

Q. Which parameters are used at microwave frequencies, and why not Z/Y?

A. S-parameters. At GHz, true opens and shorts are impossible to make (they reflect and may make active devices oscillate), so we measure with matched 50 Ω loads instead.

4.6 Conditions for reciprocity and symmetry

Reciprocal: swapping the source and the meter between the ports gives the same reading (Module 2's reciprocity theorem). Any network of R, L, C and transformers is reciprocal; dependent sources usually break it.

Derivation (Z). Drive port 1 with current I, port 2 open: $V_2 = Z_{21}I$. Now drive port 2 with the same I, port 1 open: $V_1 = Z_{12}I$. Reciprocity says these open-circuit voltages are equal ⇒ $\mathbf{Z_{12} = Z_{21}}$. The other conditions follow from the conversion formulas (Section 4.7): $Y_{12} = -Z_{12}/\Delta Z$, so $Y_{12} = Y_{21}$; $h_{12} = Z_{12}/Z_{22}$ and $h_{21} = -Z_{21}/Z_{22}$, so $h_{12} = -h_{21}$; and $AD - BC = Z_{12}/Z_{21} = 1$.

Symmetrical: the ports can be interchanged without changing port voltages and currents — the network "looks the same from both ends" ($Z_{11} = Z_{22}$). Example: a T with equal series arms.

Parameter set	Reciprocity	Symmetry (in addition to reciprocity)
Z	$Z_{12} = Z_{21}$	$Z_{11} = Z_{22}$
Y	$Y_{12} = Y_{21}$	$Y_{11} = Y_{22}$
h	$h_{12} = -h_{21}$	$\Delta h = h_{11}h_{22} - h_{12}h_{21} = 1$
ABCD	$AD - BC = 1$	$A = D$

So a reciprocal two-port needs only **3** independent numbers, and a symmetrical reciprocal one only **2**.

COMMON MISTAKE

Writing $h_{12} = h_{21}$ for reciprocity. For h parameters the condition has a **minus** sign: $h_{12} = -h_{21}$ (check Example 4.6: 0.6 and −0.6).

4.7 Interrelationships between parameters

Every set can be converted to every other (as long as the needed parameters exist). The method is always the same: take the equations of the known set and rearrange them into the form of the wanted set. You don't need to memorise all of the table below — know how to derive one row, and remember the determinants: $\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21}$, and similarly ΔY, Δh, $\Delta T = AD - BC$.

Sample derivation (Z → ABCD). From $V_2 = Z_{21}I_1 + Z_{22}I_2$: $I_1 = (1/Z_{21})V_2 - (Z_{22}/Z_{21})I_2 \Rightarrow C = 1/Z_{21}$, $D = Z_{22}/Z_{21}$. Substitute into $V_1 = Z_{11}I_1 + Z_{12}I_2$: $V_1 = (Z_{11}/Z_{21})V_2 - (\Delta Z/Z_{21})I_2 \Rightarrow A = Z_{11}/Z_{21}$, $B = \Delta Z/Z_{21}$.

Wanted	from Z	from Y	from h	from ABCD
Z	—	$Z_{11} = Y_{22}/\Delta Y$ $Z_{12} = -Y_{12}/\Delta Y$ $Z_{21} = -Y_{21}/\Delta Y$ $Z_{22} = Y_{11}/\Delta Y$	$Z_{11} = \Delta h/h_{22}$ $Z_{12} = h_{12}/h_{22}$ $Z_{21} = -h_{21}/h_{22}$ $Z_{22} = 1/h_{22}$	$Z_{11} = A/C$ $Z_{12} = \Delta T/C$ $Z_{21} = 1/C$ $Z_{22} = D/C$
Y	$Y_{11} = Z_{22}/\Delta Z$ $Y_{12} = -Z_{12}/\Delta Z$ $Y_{21} = -Z_{21}/\Delta Z$ $Y_{22} = Z_{11}/\Delta Z$	—	$Y_{11} = 1/h_{11}$ $Y_{12} = -h_{12}/h_{11}$ $Y_{21} = h_{21}/h_{11}$ $Y_{22} = \Delta h/h_{11}$	$Y_{11} = D/B$ $Y_{12} = -\Delta T/B$ $Y_{21} = -1/B$ $Y_{22} = A/B$
h	$h_{11} = \Delta Z/Z_{22}$ $h_{12} = Z_{12}/Z_{22}$ $h_{21} = -Z_{21}/Z_{22}$ $h_{22} = 1/Z_{22}$	$h_{11} = 1/Y_{11}$ $h_{12} = -Y_{12}/Y_{11}$ $h_{21} = Y_{21}/Y_{11}$ $h_{22} = \Delta Y/Y_{11}$	—	$h_{11} = B/D$ $h_{12} = \Delta T/D$ $h_{21} = -1/D$ $h_{22} = C/D$
ABCD	$A = Z_{11}/Z_{21}$ $B = \Delta Z/Z_{21}$ $C = 1/Z_{21}$ $D = Z_{22}/Z_{21}$	$A = -Y_{22}/Y_{21}$ $B = -1/Y_{21}$ $C = -\Delta Y/Y_{21}$ $D = -Y_{11}/Y_{21}$	$A = -\Delta h/h_{21}$ $B = -h_{11}/h_{21}$ $C = -h_{22}/h_{21}$ $D = -1/h_{21}$	—

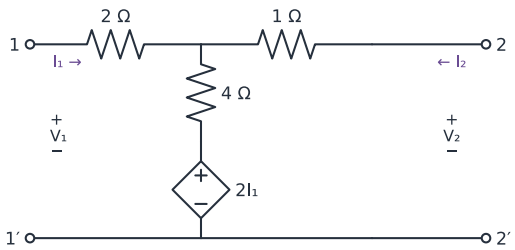


Fig 4.8 T network with a CCVS in the shunt arm (Example 4.8).

SOLVED EXAMPLE 4.8 · DEPENDENT SOURCE + CONVERSIONS

Find the Z parameters of Fig 4.8, test reciprocity, and convert to Y, h and ABCD.

- Middle node voltage: $V_x = 4(I_1 + I_2) + 2I_1 = 6I_1 + 4I_2$.
- $V_1 = 2I_1 + V_x = 8I_1 + 4I_2$; $V_2 = 1{\cdot}I_2 + V_x = 6I_1 + 5I_2 \Rightarrow [Z] = [8\ 4;\ 6\ 5]\ \Omega$.
- $Z_{12} = 4 \neq Z_{21} = 6 \Rightarrow$ **not reciprocal** (the dependent source breaks it). $\Delta Z = 40 - 24 = 16$.
- $Y = (1/\Delta Z)[Z_{22}\ -Z_{12};\ -Z_{21}\ Z_{11}] = [0.3125\ -0.25;\ -0.375\ 0.5]$ S.
- h: $h_{11} = 16/5 = 3.2\ \Omega$, $h_{12} = 4/5 = 0.8$, $h_{21} = -6/5 = -1.2$, $h_{22} = 1/5 = 0.2$ S.

▸ ABCD: A = 8/6 = 1.333, B = 16/6 = 2.667 Ω, C = 1/6 = 0.1667 S, D = 5/6 = 0.8333. AD − BC = 10/9 − 4/9 = 0.667 = Z_{12}/Z_{21} √ (≠ 1).

[Z] = [8 4; 6 5] Ω, non-reciprocal; [Y] = [0.3125 −0.25; −0.375 0.5] S; h = 3.2 Ω, 0.8, −1.2, 0.2 S; A, B, C, D = 1.333, 2.667 Ω, 0.1667 S, 0.8333

4.8 Series, parallel and cascade connections

Big networks are often built from smaller two-ports. The trick is to choose the parameter set that simply **adds** or **multiplies** for that kind of connection:

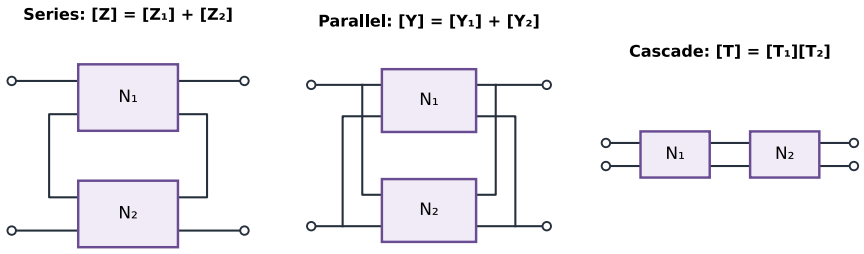


Fig 4.9 The three interconnections and the parameter set that combines simply in each.

Connection	What is common	Rule	Why
Series–series	same port currents	$[Z] = [Z_1] + [Z_2]$	port voltages add, currents equal
Parallel–parallel	same port voltages	$[Y] = [Y_1] + [Y_2]$	port currents add, voltages equal
Series–parallel	I_1 and V_2	$[h] = [h_1] + [h_2]$	(used for feedback amplifiers)
Cascade (tandem)	output of N_1 = input of N_2	$[T] = [T_1][T_2]$ (order matters!)	V_2, $-I_2$ of N_1 = V_1, I_1 of N_2

Proof for cascade. For N_1: $[V_1; I_1] = [T_1][V_a; -I_a]$, and the same pair $(V_a, -I_a)$ is the input of N_2: $[V_a; -I_a] = [T_2][V_2; -I_2]$. Substituting, $[V_1; I_1] = [T_1][T_2][V_2; -I_2]$. That is why ABCD has its strange minus sign — it makes chaining a plain matrix product.

COMMON MISTAKE

Adding Z (or Y) matrices when the connection upsets the "port condition" — the current into one terminal of a port must still equal the current out of the other. Series/parallel addition is valid when this holds (check with **Brune's test**; it always holds for three-terminal networks with a common ground line, or with ideal 1:1 isolating transformers). Cascade multiplication is always valid.

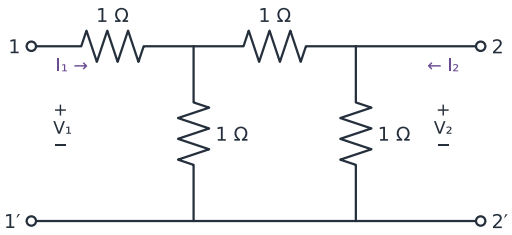


Fig 4.10 Two identical L-sections in cascade (Example 4.9).

SOLVED EXAMPLE 4.9 · CASCADE

Fig 4.10 is two identical sections, each a series 1 Ω followed by a shunt 1 Ω. Using ABCD parameters, find the overall ABCD matrix, the input resistance with port 2 open and the open-circuit voltage gain V_2/V_1.

- Series Z: [1 Z; 0 1]. Shunt Y: [1 0; Y 1]. One section: [1 1; 0 1][1 0; 1 1] = [2 1; 1 1].
- Two sections: [2 1; 1 1]·[2 1; 1 1] = [4+1 2+1; 2+1 1+1] = [5 3; 3 2]. AD − BC = 10 − 9 = 1 √.
- Port 2 open ($I_2 = 0$): $R_{in} = V_1/I_1 = A/C = 5/3 = 1.667$ Ω. Check by ladder reduction: 1 + 1 ∥ (1 + 1) = 1 + 2/3 √.
- $V_2/V_1 = 1/A = 0.2$.

[T] = [5 3; 3 2], R_{in} = 1.667 Ω, V_2/V_1 = 0.2

SOLVED EXAMPLE 4.10 · SERIES AND PARALLEL

(a) The T of Fig 4.5 is connected in series with a two-port having [Z] = [3 1; 1 2] Ω. Find the overall Z. (b) A 2 Ω resistor is connected directly between terminals 1 and 2 of the π of Fig 4.6 (a parallel connection). Find the overall Y.

- (a) [Z] = [8 6; 6 10] + [3 1; 1 2] = [11 7; 7 12] Ω.
- (b) A lone series 2 Ω has [Y] = [0.5 −0.5; −0.5 0.5] S. Add: [1.5 −1; −1 1.25] + [0.5 −0.5; −0.5 0.5] = [2 −1.5; −1.5 1.75] S.
- Sanity check (b): the 1 Ω and 2 Ω are now in parallel (0.667 Ω, 1.5 S), so Y_{12} = −1.5 S √.

(a) [Z] = [11 7; 7 12] Ω (b) [Y] = [2 −1.5; −1.5 1.75] S

✓ Module 4 — Quick revision

Network function: H(s) = response/excitation, zero ICs; h(t) = $\mathcal{L}^{-1}\{H(s)\}$.

Driving point: same port (Z_{11}, Y_{11}). **Transfer:** different ports (G_{21}, α_{21}, Z_{21}, Y_{21}).

Poles (×) = natural frequencies; **zeros (○)** = zero transmission. LHP stable, jω simple = marginal, RHP unstable.

Restrictions: DP — positive coefficients, poles & zeros in closed LHP (jω simple), degree difference ≤ 1. TF — poles in closed LHP, zeros anywhere.

Z: V = ZI, open-circuit tests. **Y:** I = YV, short-circuit tests; $Y = Z^{-1}$.

h: $V_1 = h_{11}I_1 + h_{12}V_2$, $I_2 = h_{21}I_1 + h_{22}V_2$.

ABCD: $V_1 = AV_2 - BI_2$, $I_1 = CV_2 - DI_2$.

Reciprocal: $Z_{12} = Z_{21}$, $Y_{12} = Y_{21}$, $h_{12} = -h_{21}$, AD − BC = 1. **Symmetric:** $Z_{11} = Z_{22}$, $Y_{11} = Y_{22}$, Δh = 1, A = D.

T network: $Z_{11} = Z_a + Z_c$, $Z_{22} = Z_b + Z_c$, $Z_{12} = Z_c$. **π:** $Y_{11} = Y_A + Y_C$, $Y_{12} = -Y_C$.

Connections: series → Z add; parallel → Y add; series-parallel → h add; cascade → T multiply.

Practice — short questions (3 marks)

1. Define driving-point and transfer functions with examples.
2. What is the significance of poles and zeros? Sketch the response for a pole pair at $-1 \pm j5$.
3. List the restrictions on pole and zero locations for driving-point functions.
4. Is $Z(s) = (s + 1)/(s^2 - s + 2)$ a valid driving-point impedance? Why? [No — negative coefficient, RHP poles]
5. Define h parameters and draw the h-parameter equivalent circuit.
6. Derive the condition for reciprocity in terms of Z parameters.
7. Find the Z parameters of a T with series arms 3 Ω, 3 Ω and shunt 5 Ω. Is it symmetric? [8, 5, 5, 8 Ω — yes]

Practice — long questions (9/14 marks)

1. In Fig 4.2 change the 1 Ω to 2 Ω; find $Z_{11}(s)$, V_2/V_1 and plot poles and zeros. [Ans: $(2s^2 + s + 2)/(2s + 1)$; $2/(2s^2 + s + 2)$]
2. Find the Y parameters of Fig 4.5 and verify by inverting [Z]. [Ans: $Y_{11} = 0.2273$, $Y_{12} = Y_{21} = -0.1364$, $Y_{22} = 0.1818$ S]
3. Express h parameters in terms of Z parameters and ABCD parameters in terms of Y parameters.
4. Find the Z and h parameters of Fig 4.8 if the CCVS is changed to $4I_2$. Is it reciprocal now? [Z = [6 8; 4 9], no; h = 22/9 Ω, 8/9, −4/9, 1/9 S]
5. Two identical networks with [Z] = [4 2; 2 4] Ω are connected in cascade. Find the overall ABCD and Z parameters. [Ans: T = [7 24; 2 7]; Z = [3.5 0.5; 0.5 3.5] Ω]

⚡ Placement & GATE rapid-fire

Cover the answer, say yours out loud, then check. Interviewers care more about the *reason* than the number, so practise giving the one-line "why" too. Questions marked tricky are the ones that catch people out.

1. **Three 6 Ω resistors in parallel?** 2 Ω (n equal resistors in parallel: R/n).
2. **KCL and KVL are consequences of what?** Conservation of charge and conservation of energy, valid for lumped circuits.
3. **A circuit has 7 branches and 4 nodes. How many mesh and node equations?** Mesh: b − n + 1 = 4. Node: n − 1 = 3.
4. **When do you use a supernode?** When a voltage source sits between two non-reference nodes; write one KCL for the bubble plus the source constraint.
5. **Can you source-transform an ideal voltage source?** No — it needs a series resistance; $R_s = 0$ would give an infinite current source.
6. **Two ideal current sources of 2 A and 3 A in series?** tricky Not allowed — it violates KCL. Likewise, unequal ideal voltage sources in parallel violate KVL.
7. **Can superposition find power?** No — power ∝ I^2, not linear. Find total current first. (Exception: sources at different frequencies.)
8. **How do you "switch off" sources in superposition/Thevenin?** Voltage source → short, current source → open. Dependent sources are never switched off.
9. **V_{th} = 12 V, R_{th} = 4 Ω. Norton current?** $I_N = 12/4 = 3$ A, $R_N = 4$ Ω.
10. **Thevenin equivalent of a network containing only dependent sources and resistors?** $V_{th} = 0$; R_{th} from a test source (it can even be negative).
11. **A 10 V source with 2 Ω internal resistance: maximum power to a load?** $V^2/4R = 100/8 = 12.5$ W at $R_L = 2$ Ω, efficiency 50 %.
12. **$Z_{th} = 3 + j4$ Ω and the load must be a pure resistor. Best R_L?** $|Z_{th}| = 5$ Ω. If Z_L is free: $3 - j4$ Ω (conjugate).
13. **Why don't power grids use maximum power transfer?** 50 % efficiency is unacceptable; they keep $R_{source} \ll R_{load}$. MPT is for weak signals (RF, sensors, audio).
14. **Conditions for the reciprocity theorem?** Linear, bilateral, single independent source, no dependent sources.
15. **Millman: 10 V/5 Ω and 20 V/5 Ω in parallel?** V = (2 + 4)/(0.4) = 15 V, R = 2.5 Ω.
16. **Inductor behaviour at $t = 0^+$ (uncharged) and at DC steady state?** Open circuit at first, short circuit finally. Capacitor: the opposite.
17. **Energy in 2 H carrying 3 A? In 4 µF at 10 V?** ½Li^2 = 9 J; ½CV^2 = 200 µJ.
18. **A 1 F capacitor at 2 V is connected to an identical uncharged one. Final voltage and energy?** tricky 1 V each (charge shared). Energy drops from 2 J to 1 J — half is lost in the wire resistance/radiation, however small the resistance.
19. **Time constant of R = 100 Ω, L = 10 mH?** τ = L/R = 0.1 ms. (RC: R = 1 kΩ, C = 10 µF gives 10 ms.)
20. **What fraction is reached after τ, and why do we say "settled" after 5τ?** 63.2 %; after 5τ, 99.3 % (within 1 %).
21. **An ideal inductor is connected to a DC voltage V at t = 0. Current?** i = (V/L)t — a ramp forever (no resistance to limit it).
22. **Can an impulse change a capacitor voltage instantly?** Yes — an impulse current delivers finite charge in zero time. Only finite inputs obey the "no jump" rule.
23. **$\mathcal{L}\{\delta(t-2)\}$? $\mathcal{L}\{t\,e^{-3t}\}$?** e^{-2s}; $1/(s+3)^2$.
24. **Final value of f(t) for $F(s) = 1/[s(s+2)]$? For $F(s) = 1/(s^2+4)$?** tricky 0.5. For the second, FVT is not valid — poles on the jω-axis, f(t) oscillates.
25. **Laplace vs Fourier?** Laplace uses $s = \sigma + j\omega$, handles growing signals, transients and initial conditions; Fourier ($s = j\omega$) gives steady-state frequency response.
26. **Series RLC: L = 1 mH, C = 1 µF. Resonant frequency and critical R?** $\omega_0 = 31.62$ krad/s ($f_0 = 5.03$ kHz); $R_{crit} = 2\sqrt{(L/C)} = 63.25$ Ω.

27. **Same circuit with R = 10 Ω: Q and damping?** Q = $(1/R)\sqrt{(L/C)}$ = 3.16, $\zeta = 1/2Q = 0.158$ → underdamped, rings.
28. **In a parallel RLC, does increasing R increase damping?** tricky No — $\alpha = 1/2RC$, so larger R means less damping (opposite of series RLC).
29. **RC low-pass with f_c = 1 MHz: rise time?** $t_r = 0.35/f_c$ = 0.35 µs.
30. **When is an RC high-pass a differentiator? A low-pass an integrator?** HPF: $\tau \ll$ pulse width/period. LPF: $\tau \gg$ pulse width/period.
31. **Impulse response of H(s) = 1/(s + 3)? Its step response?** e^{-3t}; $(1 - e^{-3t})/3$.
32. **What do poles and zeros physically represent?** Poles: natural frequencies (modes of the free response). Zeros: frequencies of zero transmission.
33. **A zero at s = 0 in a transfer function means?** The network blocks DC — high-pass behaviour (a series capacitor somewhere).
34. **Poles at ±j5 (simple)? Repeated poles at ±j5?** Simple: sustained oscillation, marginally stable. Repeated: t·sin 5t — unstable.
35. **Can a passive driving-point impedance have a zero in the RHP?** No. Transfer functions can (non-minimum-phase), driving-point functions cannot.
36. **Maximum degree difference between N(s) and D(s) for a driving-point function?** 1 (at extreme frequencies it must look like R, L or C).
37. **Why are they called "open-circuit" Z parameters?** Each is measured with the other port open (its current zero).
38. **Sign of Y_{12} for a passive π network?** Negative: $-Y_{series}$, because I_2 is defined into the port.
39. **Reciprocity condition in h and ABCD?** $h_{12} = -h_{21}$; AD − BC = 1.
40. **Symmetry condition in ABCD and h?** A = D; $\Delta h = 1$.
41. **How many independent parameters: reciprocal two-port? reciprocal + symmetric?** 3; 2.
42. **ABCD of a series impedance Z? Of a shunt admittance Y?** [1 Z; 0 1] and [1 0; Y 1].
43. **Which parameters add for series, parallel and series-parallel connections; which multiply for cascade?** Z, Y, h add; ABCD multiply (in order).
44. **Do Z parameters exist for an ideal transformer?** tricky No (nor Y). ABCD exists: [n 0; 0 1/n].
45. **ABCD of a lossless transmission line of length l?** A = D = cos βl, B = jZ_0 sin βl, C = j sin βl / Z_0.
46. **Why h parameters for BJTs, S parameters for RF?** h tests (output short, input open) suit the BJT's low input / high output impedance; at RF ideal opens/shorts are impossible, so matched loads (S) are used.

Σ Formula sheet

Module 1 — Basics, mesh and node

Item	Formula
Element laws	$v = Ri$; $v = L\,di/dt$; $i = C\,dv/dt$; $w_L = \frac{1}{2}Li^2$, $w_C = \frac{1}{2}Cv^2$
Practical sources	$V = V_s - IR_s$; $I = I_s - V/R_p$
Source transformation	V_s in series with $R \Leftrightarrow I_s = V_s/R$ in parallel with R
Dividers	$V_k = V \cdot R_k/\Sigma R$ (series); $I_1 = I \cdot R_2/(R_1 + R_2)$ (two in parallel)
Number of equations	mesh: $b - n + 1$; node: $n - 1$
Mesh by inspection	$[R][I] = [V]$: R_{kk} = self, R_{kj} = −shared (independent V sources)
Node by inspection	$[G][V] = [I]$: G_{kk} = self, G_{kj} = −mutual (independent I sources)
AC	$Z_L = j\omega L$, $Z_C = -j/\omega C$; $P = \lvert V\rvert\lvert I\rvert \cos\theta$ (rms) = $\mathrm{Re}(VI^*)$

Module 2 — Theorems

Theorem	Key relation
Thevenin / Norton	$V_{th} = V_{oc}$; $I_N = I_{sc}$; $R_{th} = V_{oc}/I_{sc} = V_{test}/I_{test}$; $I_L = V_{th}/(R_{th} + R_L)$
Max power (DC)	$R_L = R_{th}$, $P_{max} = V_{th}^2/4R_{th}$, $\eta = 50\ \%$
Max power (AC)	$Z_L = Z_{th}^*$: $P_{max} = \lvert V_{th}\rvert^2/4R_{th}$; pure R_L: $R_L = \lvert Z_{th}\rvert$
Millman	$V = \Sigma E_k G_k/\Sigma G_k$, $R = 1/\Sigma G_k$
Reciprocity	E/I ratio unchanged when source and meter are interchanged

Module 3 — Laplace and transients

Item	Formula
Pairs	$\delta \to 1$; $u \to 1/s$; $t^n \to n!/s^{n+1}$; $e^{-at} \to 1/(s+a)$; $\sin\omega t \to \omega/(s^2+\omega^2)$; $\cos\omega t \to s/(s^2+\omega^2)$
Damped	$e^{-at}\sin\omega t \to \omega/[(s+a)^2+\omega^2]$; $e^{-at}\cos\omega t \to (s+a)/[(s+a)^2+\omega^2]$; $te^{-at} \to 1/(s+a)^2$
Properties	$f(t-T)u(t-T) \to e^{-sT}F$; $e^{-at}f \to F(s+a)$; $f' \to sF - f(0^-)$; $\int f \to F/s$; $tf \to -dF/ds$
IVT / FVT	$f(0^+) = \lim_{s\to\infty} sF(s)$; $f(\infty) = \lim_{s\to 0} sF(s)$ (poles of sF in LHP)
s-domain elements	sL with series source $Li(0^-)$ (or ∥ $i(0^-)/s$); $1/sC$ with series $v(0^-)/s$ (or ∥ $Cv(0^-)$)
First order	$x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$; $\tau = R_{th}C$ or L/R_{th}
Series RLC	$s^2 + (R/L)s + 1/LC = 0$; $\alpha = R/2L$, $\omega_0 = 1/\sqrt{LC}$, $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$; $R_{crit} = 2\sqrt{(L/C)}$; $Q = \omega_0 L/R$
Parallel RLC	$\alpha = 1/2RC$, same ω_0
RC filters	$H_{LP} = 1/(1+sRC)$, $H_{HP} = sRC/(1+sRC)$; $f_c = 1/2\pi RC$; $t_r = 2.2RC = 0.35/f_c$

Module 4 — Network functions and two-ports

Set	Equations	Reciprocal	Symmetric
Z (open circuit)	$V_1 = Z_{11}I_1 + Z_{12}I_2$; $V_2 = Z_{21}I_1 + Z_{22}I_2$	$Z_{12} = Z_{21}$	$Z_{11} = Z_{22}$
Y (short circuit)	$I_1 = Y_{11}V_1 + Y_{12}V_2$; $I_2 = Y_{21}V_1 + Y_{22}V_2$	$Y_{12} = Y_{21}$	$Y_{11} = Y_{22}$
h (hybrid)	$V_1 = h_{11}I_1 + h_{12}V_2$; $I_2 = h_{21}I_1 + h_{22}V_2$	$h_{12} = -h_{21}$	$\Delta h = 1$
ABCD (transmission)	$V_1 = AV_2 - BI_2$; $I_1 = CV_2 - DI_2$	$AD - BC = 1$	$A = D$

Item	Formula
Most-used conversions	$[Y] = [Z]^{-1}$; $h_{11} = \Delta Z/Z_{22}$, $h_{12} = Z_{12}/Z_{22}$, $h_{21} = -Z_{21}/Z_{22}$, $h_{22} = 1/Z_{22}$; $A = Z_{11}/Z_{21}$, $B = \Delta Z/Z_{21}$, $C = 1/Z_{21}$, $D = Z_{22}/Z_{21}$; $Z_{11} = A/C$, $Z_{12} = \Delta T/C$, $Z_{21} = 1/C$, $Z_{22} = D/C$
T and π	T: $Z_{11} = Z_a + Z_c$, $Z_{22} = Z_b + Z_c$, $Z_{12} = Z_c$. π: $Y_{11} = Y_A + Y_C$, $Y_{22} = Y_B + Y_C$, $Y_{12} = -Y_C$
Interconnections	series: Z add; parallel: Y add; series-parallel: h add; cascade: $[T] = [T_1][T_2]$
Elementary ABCD	series Z: [1 Z; 0 1]; shunt Y: [1 0; Y 1]; ideal transformer n:1: [n 0; 0 1/n]
Stability from poles	LHP: stable; simple jω: marginal; RHP or repeated jω: unstable

☾ Night before the exam

Breathe. You don't need to re-read the book tonight — just make sure these ideas feel familiar. If you can explain each line below to a friend, you are ready. Then sleep well: a rested brain solves circuits faster. All the best!

MODULE 1 — FIVE THINGS

1. Passive sign convention: current into + means power absorbed. Check $\Sigma p = 0$.
2. Mesh: KVL per window, mutual terms negative. Node: KCL per node, choose ground wisely.
3. Current source in a shared branch → supermesh. Voltage source between nodes → supernode.
4. Dependent source: treat as normal, then add one equation for its controlling variable.
5. AC: same methods with impedances; subtract phasors in rectangular form.

MODULE 2 — FIVE THINGS

1. Superposition: one independent source at a time; never for power; dependent sources stay.
2. Thevenin: V_{oc}, then R_{th} (kill sources, or V_{oc}/I_{sc}, or test source when dependent sources exist).
3. MPT: $R_L = R_{th}$, $P = V^2/4R$; AC: conjugate match.
4. Reciprocity needs a single source and no dependent sources.
5. Millman = node equation in disguise: $V = \Sigma EG/\Sigma G$.

MODULE 3 — FIVE THINGS

1. Table of transforms + shifts + derivative rule ($sF - f(0^-)$).
2. Partial fractions: cover-up; repeated → differentiate; complex → complete the square.
3. Initial conditions become sources: $Li(0^-)$ and $v(0^-)/s$.
4. First order: $x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$. Second order: compare R with $2\sqrt{(L/C)}$.
5. RC LPF/HPF pulse sketches for $\tau \ll t_p$ and $\tau \gg t_p$; HPF goes negative after the pulse.

MODULE 4 — FIVE THINGS

1. Driving point = same port; transfer = different ports.
2. Poles × decide response shape and stability; zeros ○ decide amplitudes.
3. Driving-point rules: positive coefficients, nothing in the RHP, degree difference ≤ 1.
4. Z (open), Y (short), h (mixed), ABCD (input vs output) — know each definition with its test condition.
5. Reciprocal: $Z_{12} = Z_{21}$, $h_{12} = -h_{21}$, $AD - BC = 1$. Connections: Z add, Y add, T multiply.

IN THE EXAM HALL

- Read all questions first; start with the one you are surest of.
- Draw the circuit again with your own labels, polarities and current directions — examiners give marks for it.
- Write the method's name and the statement of the theorem before solving; box the final answer with units.
- Cross-check one answer per question (power balance, IVT at $t = 0$, $AD - BC = 1$). It takes a minute and catches most slips.